

Asymptotics of higher rank $6j$ symbols and tetrahedral Horn problems

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March 30, 2025

Abstract

For a compact reductive group G , Wigner $6j$ -symbols are maps between multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\gamma \cong V_\alpha \otimes (V_\beta \otimes V_\gamma)$ for unitary irreducible representations of G . For $G = SU(2)$, their asymptotic behavior for large weights was conjectured by Ponzano-Regge and established by Roberts using geometric quantization techniques.

In this paper, we address the problem of asymptotic behavior of $6j$ -symbols for $G = U(n)$. We show that their decay rate is polynomial or exponential depending on the existence of a tetrahedron in the space of Hermitian n by n matrices with edges determined by the 6 arguments of the $6j$ -symbol. Our proof is inspired by probabilistic techniques from quantum information theory.

As an application of our results, we obtain a new proof of the strongly subadditive entropy inequality for tetrahedra in the space of Hermitian matrices (the tetrahedral Horn problem).

Wigner $6j$ -symbols play a fundamental role in quantum physics displaying $SU(2)$ symmetry. The semiclassical behaviour or the $6j$ -symbols, as conjectured by Ponzano and Regge and later proven using geometric quantization techniques by Roberts, is intimately related to the existence of Euclidean tetrahedra whose sidelengths correspond to the six js . Upon generalizing to $SU(n)$ for $n \geq 3$, the corresponding $6j$ symbol becomes an operator and, although still a central object in quantum theory, results concerning its semiclassical asymptotics are lacking. Here, we connect the asymptotics of (the norm of) the higher rank $6j$ symbols to the solution of a tetrahedral generalization of Horn's problem concerning sums of n -by- n Hermitian matrices. Our result is consistent with the special case where (the norm of) the $6j$ symbol becomes the Littlewood-Richardson coefficient, whose semiclassical limit corresponds to solutions to Horn's problem. Our approach is through Schur-Weyl duality and the method of eigenvalue estimation previously developed in the context of quantum information theory. As a corollary, we provide a new perspective on a strongly subadditive unnormalized entropy inequality constraining tetrahedral eigenvalues.

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1 Introduction

[OUTLINE: What are $6j$ Symbols?]

[OUTLINE: What has been done for $SU(2)$, Wigner, Neville, Ponzano & Regge, Roberts, Barrett-Crane type state-sum models, loop quantum gravity, etc...]

Various $k \rightarrow \infty$ asymptotics for the $U_q(\mathfrak{sl}_2)$ $6j$ symbol were calculated by Taylor and Woodward and shown to encode (i) the existence a spherical tetrahedron on the unit sphere with prescribed edgelengths when $q = \exp(i\pi/r(k))$ is a root of unity (with $r(k) = k(r-2) + 2$) and (ii) the existence of a hyperbolic tetrahedron with prescribed edgelengths when $q = \exp(\pi/r(k))$ [TW06].

See also the thesis of Backens [Bac10].

[OUTLINE: Previous work on $SU(N)$]

Over the years, much attention placed on the study of $SU(n)$ or $U_q(\mathfrak{sl}_n)$ $6j$ symbols has been on the multiplicity-free cases where the dimensions of each of the multiplicity spaces is either one or zero (in the latter case the $6j$ symbol is zero).

For example, tensor products between symmetric representations $\lambda = (k)$ or conjugates of symmetric representations $\lambda = (k^{(n-1)})^1$ are necessarily multiplicity free; Nawata, Pichai, and Zodinmawia conjectured closed form expressions for the $U_q(\mathfrak{sl}_n)$ $6j$ symbols when a, b, d and e are integers and either $(\alpha, \beta, \delta, \epsilon) = ((a), (b^{(n-1)}), (d), (e^{(n-1)}))$ and $a + d = b + e$ (Type I) or $(\alpha, \beta, \delta, \epsilon) = ((a), (b), (d^{(n-1)}), (e^{(n-1)}))$ and $a + b = d + e$ (Type II) as well as the cases which can be related to these by known symmetries of the corresponding $6j$ symbol [NPZ13]. Building upon this work, Alekseev, Morozov, and Sleptsov studied both the asymptotics of and symmetries in each of the Type I and Type II cases [AMS20].

More recently, Alcock-Zeilinger et al. have derived formulae for the family of $SU(n)$ $6j$ symbols where at least one pair of non-adjacent edges, say (α, δ) correspond to a pair of fundamental/defining representations, i.e., $\alpha, \delta = (1)$ [Alc+23]. Shortly thereafter, Keppeler, Plätzer, and Sjødahl expanded upon these results to derive formulae for the family of $SU(n)$ $6j$ symbols where either (i) at least one pair of non-adjacent edges consists of fundamental or adjoint representations, i.e., $\alpha, \delta \in \{(1), (2, 1^{(n-2)})\}$, or (ii) at least one vertex, say (α, β, ϕ) , consists entirely of fundamental or adjoint representations, i.e., $\alpha, \beta, \phi \in \{(1), (2, 1^{(n-2)})\}$ [KPS24].

TC: Here are some research notes concerning the $6j$ symbols:

- For a comprehensive reference textbook for the $6j$ coefficients, see the textbook by Varshalovich, Moskalev, and Khersonskii [VMK88], specifically Chapter 9.
- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- Representation theoretic proof of the Regge symmetries due to Boalch [Boa07]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].

¹Here $k^{(n-1)}$ is shorthand for a string $k, k, \dots, k$ consisting of $n - 1$ repetitions of k .

- Only the tetrahedral symmetry of the $U_q(SL(N))$ $6j$ -symbol is known, the counterpart of the Regge symmetry is still unknown, although Morozov and Sleptsov conjectures the generalized Regge symmetry may be related to the eigenvalue conjecture, just as it is in the case when $N = 2$ [MS18].
- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- The original Ponzano-Regge paper [PR68]
- Gurau derives the Ponzano-Regge asymptotic formula directly using Stirling's approximation applied to Racah's single-sum formula [Gur08]
- Bartlett and Nzaganya work with so-called *coherent loop states* or *Bohr-Sommerfeld states* to develop some semiclassical results concerning the inner products between such states (see also the talk on Youtube) [BN23]
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- Marzuoli and Rasetti connects quantum computing models to the recoupling theory of $SU(2)$ angular momentum and also considers the semiclassical limit [MR05]
- Artamonov derives an explicit formula for the $n = 3$ case [Art23], and later to arbitrary n [Art24].
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension n but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all n [LS23].

[OUTLINE: What we do for $SU(N)$]

The purpose of this paper is to report the following pair of complementary bounds on the norm $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ of the $6j$ symbols for the unitary group $U(n)$.

Theorem 1. *If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are the eigenvalues of $n \times n$ positive semidefinite matrices (A, B, C, D, E, F) satisfying*

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad \text{eq:tet_cond} \quad (1)$$

then there exists a sequence of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ satisfying

$$\frac{\alpha'_k}{k} \rightarrow \alpha, \quad \frac{\beta'_k}{k} \rightarrow \beta, \quad \frac{\gamma'_k}{k} \rightarrow \gamma, \quad \frac{\delta'_k}{k} \rightarrow \delta, \quad \frac{\epsilon'_k}{k} \rightarrow \epsilon, \quad \frac{\phi'_k}{k} \rightarrow \phi, \quad \text{eq:six_convergence} \quad (2)$$

and

$$\left\| \begin{Bmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{Bmatrix} \right\| \geq \frac{1}{\text{poly}(k)}. \quad (3)$$

Otherwise, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not the eigenvalues of any matrices satisfying Eq. (1), then for any sequence of Young diagrams satisfying Eq. (2), we have

$$\left\| \begin{Bmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{Bmatrix} \right\| \leq \exp(-\Omega(k)). \quad (4)$$

[OUTLINE: Horn's problem]

The problem of determining the set of eigenvalues (α, β, γ) of Hermitian matrices (A, B, C) satisfying $A+B=C$ has been investigated for over a century [Wey12; Fan49; Lid53; Wie55; TF71; TF70] and has come to be known as *Horn's problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. In 1998, Klyachko provided a solution

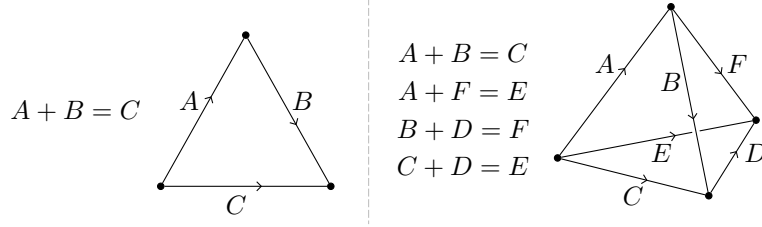


Figure 1: (Left) The *triangular* Horn's problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn's problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. [fig:tetrahedron_labelling_scheme](#)

to Horn's problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn's original conjecture [Ful98]. These linear inequalities, now known as the *Horn inequalities* describe the convex polytope of admissible values for (α, β, γ) . Recent developments on Horn's problem include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues γ of the sum $C = A + B$ if A and B are sampled uniformly and independently from the respective sets of Hermitian matrices with eigenvalues α and β by Coquereaux, McSwiggen, and Zuber in series of papers [CZ18; CZ18; CMZ19; Zub18]. For a pedagogical and historical introduction to Horn's problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00] or alternatively [NP06, Appendix D]. See also the article by Daftuar and Hayden for an account of applications to topics in quantum information theory [DH05].

In parallel to Horn's problem, numerous variations and extensions of Horn's problem have been explored by various authors. For example, numerous authors have considered the admissible eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00; AMW01].

In this paper, we consider a generalization of Horn's problem investigated previously only briefly in [CSW18]. While Horn's problem aims to characterize the eigenvalues (α, β, γ) of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization we consider here aims to characterize the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints: $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. These constraints arise naturally from building a tetrahedral arrangement of $n \times n$ Hermitian matrices inside the vector space of such Hermitian matrices with edges labelled according to Fig. 1.

Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of 2×2 Hermitian matrices, this connection is much deeper. It is well-known that the space of admissible eigenvalues for a triangular arrangement of 2×2 traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. This correspondence generalizes to the tetrahedral case in the sense that the admissible eigenvalues for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. While this latter result is straightforward to prove and also well-known, indeed it forms the basis of the semiclassical asymptotics of the $SU(2)$ Wigner $6j$ symbols [Rob99], it leads to a complete solution to the tetrahedral Horn's problem for 2×2 matrices; specifically, the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is provided by the positivity of so-called Cayley-Menger determinants which are polynomial inequalities in the side-lengths (a, b, c, d, e, f) [SJS18]. Even for $n = 2$, we remark that there is a strong qualitative difference between the tetrahedral Horn problem and the original Horn problem. In particular, while the Horn's problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn's polytope*), the solution space for the tetrahedral Horn's problem is not convex nor a polytope.

More recently, Alekseev et al. have investigated a few variants of the tetrahedral Horn problem concerning either the singular values of products of invertible matrices or the tropical singular values of concatenations of weighted planar networks [Ale+25]. These variants are termed *multiple* Horn problems in [Ale+25]; the tetrahedral Horn problem we consider here is termed the *additive* multiple Horn problem in the language of [Ale+25].

TODO: Talk about Horn's problem & Schubert calculus?

[OUTLINE: Quantum marginals problem]

The techniques used in this paper to derive bounds on the $6j$ symbol for the unitary group $U(n)$ are closely related to the techniques used in an earlier paper by Christandl, Şahinoğlu, and Walter which derived analogous bounds on the $6j$ symbol for the symmetric group S_k [CŞW18]. In that paper, the authors considered the problem of determining whether there exists a density matrix ρ_{ABC} with partial traces $\rho_{AB}, \rho_{BC}, \rho_A, \rho_B$, and ρ_C with a given sextuple of eigenvalues [CŞW18]. Such problems arise naturally in the context of quantum information theory and are typically referred to as quantum marginals problems [CŞW18; Fra22]. In particular, they proved the $6j$ symbols for the symmetric group S_k (referred to as the recoupling coefficients) in the limit as $k \rightarrow \infty$ either decay exponentially or can be bounded from below by a polynomial factor depending on whether there exists a density matrix with the prescribed marginal spectra [CŞW18]. Moreover, they also demonstrated how the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem and, in a complementary manner, furthermore demonstrated how, by restricting to the subgroup $S_k \subset U(k)$ of permutation matrices and applying Nishiyama's restriction theorem [Nis00], the $6j$ symbols for unitary group $U(n)$ may be viewed as a special case of the $6j$ symbols for the symmetric group S_k [CŞW18]. This latter result obtained by [CŞW18] is a generalizes how the Littlewood-Richardson coefficients can be cast as a special case of the Kronecker coefficients [Lit58; Mur55; IP23]. The approach taken by this paper differs from that of [CŞW18] in that we prove the correspondence between solutions to the tetrahedral Horn problem on one hand and the asymptotics of the $6j$ symbol for $U(n)$ on the other hand *directly*, rather than first pass through the quantum marginals problem or through the $6j$ symbols for the symmetric group.

[OUTLINE: Structure of the paper]

TODO: Descide on the structure of the paper? Finally, we prove an upper bound on general $6j$ symbols using the Cauchy-Schwarz inequality which, in the $n \rightarrow \infty$ asymptotic limit, gives rise to a novel proof of a known strongly subadditive inequality involving unnormalized entropies.

2 Tetrahedral eigenvalues

sec:tet_eig

2.1 The tetrahedral Horn problem

prob:tri_problem

Problem 2 (Trigonal Horn problem). A triple $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple (A, B, C) of $d \times d$ Hermitian matrices satisfying (i) the linear constraint

$$A + B = C, \tag{5}$$

{eq:tri_linear_constraints}

and (ii) the three eigenvalue constraints

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma. \tag{6}$$

{eq:tri_eigenvalue_constraints}

The problem of determining the set of trigonal eigenvalues (α, β, γ) is known as *Horn problem*.

prob:tet_problem

Problem 3 (Tetrahedral Horn problem). A sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in (\mathbb{R}_{\text{dec}}^n)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $n \times n$ Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \tag{7}$$

{eq:tet_linear_constraints}

and (ii) the six eigenvalue constraints

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma, \quad \text{eig}(D) = \delta, \quad \text{eig}(E) = \epsilon, \quad \text{eig}(F) = \phi. \tag{8}$$

{eq:tet_eigenvalue_constraints}

The problem of determining the set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ will be referred to as the *tetrahedral Horn problem*.

The tetrahedral Horn problem can be viewed as a generalization of the Horn's problem in a variety of ways. For instance, a triplet (α, β, γ) of eigenvalues is trigonal if and only if the sextuple $(\alpha, \beta, \gamma, 0, \gamma, \beta)$ is tetrahedral. More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four separate instances of Horn's original problem subject to additional matching constraints. Specifically, a sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is tetrahedral only if each of the following four triplets are trigonal:

$$(\alpha, \beta, \gamma), \quad (\beta, \delta, \phi), \quad (\gamma, \delta, \epsilon), \quad (\alpha, \phi, \epsilon). \quad (\text{eq:four_faces}) \quad (9)$$

If each of the above triplets are trigonal, then there exists a set of twelve Hermitian matrices with the corresponding eigenvalues satisfying

$$A + B = C, \quad B' + D = F, \quad C' + D' = E, \quad A' + F' = E'. \quad (\text{eq:four_faces_matrices}) \quad (10)$$

The tetrahedral Horn problem goes beyond four separate instances of Horn's original problem in that it additionally requires the primed and unprimed Hermitian matrices to coincide:²

$$A = A', \quad B = B', \quad C = C', \quad D = D', \quad E = E', \quad F = F'. \quad (11)$$

2.2 Low dimensional solutions

Remark 4 (Traces are tetrahedral). In the original version of Horn's problem, the eigenvalues (α, β, γ) of Hermitian matrices (A, B, C) satisfying $A + B = C$ necessarily must satisfy a basic equality condition known as the *trace condition*:

$$|\alpha| + |\beta| = |\gamma|, \quad (\text{eq:tri_trace_cond}) \quad (12)$$

where $|\lambda| := \sum_i \lambda_i$ is shorthand for the sum of the eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$. This condition must hold because $A + B = C$ implies $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ and $\text{Tr}(X) = |\lambda|$ for any Hermitian matrix X with eigenvalues λ . For the tetrahedral Horn's problem, a set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ necessarily satisfy four *trace conditions*:

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\delta| + |\gamma| = |\epsilon|, \quad |\alpha| + |\phi| = |\epsilon|. \quad (\text{eq:tet_trace_conds}) \quad (13)$$

Furthermore, for the trivial case where $n = 1$, the above trace constraints are both necessary and sufficient for the tetrahedral Horn problem.

exam:d_2

Example 5 (Tetrahedral Horn's problem for $d = 2$). Here we articulate the solution to the tetrahedral Horn's problem for 2×2 Hermitian matrices. We restrict our focus to the case of *traceless* 2×2 Hermitian matrices³. In this context, the sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ can be identified with a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that

$$\alpha = (a, -a), \quad \beta = (b, -b), \quad \gamma = (c, -c), \quad \delta = (d, -d), \quad \epsilon = (e, -e), \quad \phi = (f, -f). \quad (\text{eq:traceless_two_dim_tetra}) \quad (14)$$

Moreover, each traceless 2×2 Hermitian matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis $\{\sigma_1, \sigma_2, \sigma_3\}$ according to the formula:

$$X = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (15)$$

Under this identification, the eigenvalues of X are determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (16)$$

²Note that by conjugating the third and fourth conditions in Eq. (10) by unitaries U and V respectively, it is possible to set $C' \mapsto U^{-1}C'U = C$ and $F' \mapsto V^{-1}F'V = F$, and thus one could assume without loss of generality that $C = C'$ and $F = F'$.

³In Section 2.3, Corollary 10, we will see that this restriction can be made without loss of generality.

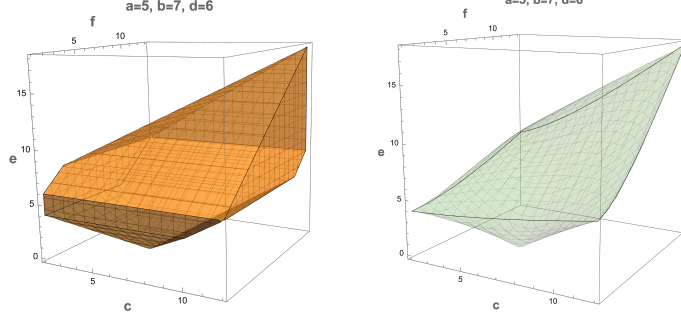


Figure 2: A slice of the set of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (17). (Right) The set of all values (c, e, f) satisfying both Eq. (17) and Eq. (18).

Furthermore, if A is determined by the coordinates (a_1, a_2, a_3) and B by the coordinates (b_1, b_2, b_3) , then their sum $C = A + B$ is determined by the coordinates $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. Consequently, in this setting, the tetrahedral Horn's problem reduces to problem of describing the space of all possible edge-lengths (a, b, c, d, e, f) of a Euclidean tetrahedron in $\mathbb{R}^3$. This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (17)$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 & e^2 \\ 1 & a^2 & 0 & b^2 & f^2 \\ 1 & c^2 & b^2 & 0 & d^2 \\ 1 & e^2 & f^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (18)$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when $D(a, b, c, d, e, f)$ is positive and when the triangle inequalities above are also satisfied, $D(a, b, c, d, e, f)$ is always equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]⁴. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (17) and Eq. (18).

2.3 Symmetries of tetrahedral eigenvalues

sec:symmetries

There are numerous types of symmetries present in the statement of Problem 3. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Problem 3 as stated.

prop:scaling

Proposition 6 (Scaling symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, $(s\alpha, s\beta, s\gamma, s\delta, s\epsilon, s\phi)$, are tetrahedral if and only if the original, unscaled eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.*

⁴The triangle inequalities $|a - b| \leq c \leq a + b$ are also equivalent to the non-negativity of the *area*, $D(a, b, c)$, of the triangle with side-lengths a, b and c where $D(a, b, c)^2 = -\frac{1}{16} \det \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & a^2 \\ 1 & a^2 & 0 \end{pmatrix}$ is equivalent to Heron's formula.

One may wonder whether it is possible to extend the scaling symmetry to consider scaling by negative $s < 0$. Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrices both negates and reverses the order of its eigenvalues.

prop:shifting

Proposition 9 (Shifting symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Given any list of non-increasing eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda + x := (\lambda_1 + x \geq \dots \geq \lambda_d + x)$. Then define shifted eigenvalues by the formula*

$$\begin{aligned} \tilde{\alpha} &:= \alpha + x, & \tilde{\beta} &:= \beta + y, & \tilde{\gamma} &:= \gamma + x + y, & \text{\textit{eq:shifted_tet_eigenvalues}} \\ \tilde{\delta} &:= \delta + z, & \tilde{\epsilon} &:= \epsilon + x + y + z, & \tilde{\phi} &:= \phi + y + z. \end{aligned} \quad (22)$$

The shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ are tetrahedral if and only if the unshifted eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.

Proof. Given a Hermitian matrix X with eigenvalues $\text{eig}(X) = \lambda$ and a constant $x \in \mathbb{R}$, the eigenvalues of $X + xI$ (where I is the identity matrix) are

$$\text{eig}(X + xI) = \lambda + x = (\lambda_1 + x \geq \dots \geq \lambda_d + x). \quad (23)$$

The statement then follows because

$$(A + xI, B + yI, C + (x + y)I, D + zI, E + (x + y + z)I, F + (y + z)I) \quad (24)$$

satisfy the tetrahedral constraints in Eq. (7) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (7) $\square$

As a corollary of the shifting symmetries (Proposition 9) one concludes that to solve the tetrahedral Horn's problem, in a certain sense, it suffices to solve the tetrahedral Horn's problem merely for traceless matrices (Corollary 10) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 11). *cor:traceless*

Corollary 10 (Traceless eigenvalues). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and for any list of n eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda^0 := \lambda - |\lambda|/n$ denote the traceless list of eigenvalues obtained by shifting λ by $-|\lambda|/n$. Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if (i) their traces $(|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|)$ are tetrahedral (they satisfy Eq. (13)) and (ii) $(\alpha^0, \beta^0, \gamma^0, \delta^0, \epsilon^0, \phi^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (13)) and applications of the shifting symmetry Proposition 9 with (x, y, z) values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \quad (25)$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn's problem it suffices to solve the tetrahedral Horn's problem for positive definite spectra only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 9 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure strictly positive eigenvalues for all six Hermitian matrices. Note that in subsequent sections we will often explicitly assume the candidate eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are either non-negative or positive for the sake of technical convenience or sometimes to establish a connection with density matrices from quantum theory (which are positive semidefinite matrices with unit trace).

cor:psd

Corollary 11 (Positive eigenvalues). *If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$\gamma_d \geq \alpha_d + \beta_d, \quad \phi_d \geq \beta_d + \delta_d, \quad \epsilon_d \geq \delta_d + \gamma_d, \quad \epsilon_d \geq \alpha_d + \phi_d. \quad (26) \quad \text{\textit{eq:min_eig_ineqs}}$$

Moreover, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a sextuple of candidate eigenvalues satisfying Eq. (26), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying

$$x > -\alpha_d, \quad y > -\beta_d, \quad z > -\delta_d, \quad (27) \quad \text{\textit{eq:positive_shift}}$$

the resulting shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ (given by Eq. (22)) are all positive.

Proof. That tetrahedral eigenvalues satisfy Eq. (26) follows from four applications of the well-known inequality $\gamma_d \geq \alpha_d + \beta_n$ for Horn's problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue λ_n of a matrix X is characterized by a minimization: $\lambda_d = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$\gamma_d = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = \alpha_d + \beta_d. \quad (28)$$

Four applications of this inequality produces Eq. (26). Now if (x, y, z) satisfy Eq. (27), we have for all j

$$\alpha_j + x \geq \alpha_d + x > 0, \quad (29)$$

meaning $\alpha + x = (\alpha_1 + x \geq \dots \geq \alpha_d + x)$ is strictly positive. Similar statements hold for $\beta, \gamma, \delta, \epsilon$ and ϕ ; for instance, for all j ,

$$\epsilon_j + x + y + z \geq \epsilon_d + x + y + z > \epsilon_d - \alpha_d - \beta_d - \delta_d \geq 0, \quad (30)$$

meaning $\epsilon + x + y + z$ is strictly positive also. $\square$

Remark 12. Although this section listed a handful of symmetries of the tetrahedral Horn's problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, e, d, f) of tetrahedra, known as the *Regge symmetries*, hold [AI19; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (31)$$

Combining this symmetry with the permutational symmetries given by Proposition 8 (and noting that for the eigenvalues λ of traceless 2×2 Hermitian matrices, $\lambda = (x, -x)$, we have $\lambda^* = \lambda$) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{aligned} (a, b, c, d, e, f) & \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & \quad (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & \quad (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned} \quad (32)$$

where

$$s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}. \quad (33)$$

Remark 13. The Regge symmetry of a tetrahedron can be understood as a consequence of an earlier discovery of a suprising symmetry of Wigner $6j$ symbol for $SU(2)$ originally due to Regge [Reg59] (see also [Boa07]). That the Regge symmetry of a tetrahedron emerges from the Regge symmetry of the associated $6j$ symbols can be viewed as a consequence of the semiclassical limit of the latter [PR68; Rob99]. It has also be shown that the Regge symmetry of Euclidean tetrahedra is actually a scissors-congruence preserving both the volume and Dhen invariants [AI19].

At the moment, it remains unclear if there exists analogous symmetries for the generalized Wigner $6j$ symbol for $n \geq 3$ (although there are conjectured symmetries for the $6j$ symbols for $U_q(\mathfrak{sl}_n)$ which would follow from the eigenvalue hypothesis [MS18; LS23]). It could also be possible for additional symmetries to be present in the tetrahedral Horn problem for $n \geq 3$ which are *not* present in the $6j$ symbol for $SU(n)$.

3 Representation theory & eigenvalue estimation

sec:rep_theory

3.1 Schur-Weyl duality & eigenvalue estimation

Our main tool for connecting the eigenvalues of an $n \times n$ Hermitian matrix A to the representation theory of the unitary group $U(n)$ will be to examine the Schur-Weyl decomposition of the k -th tensor power $A^{\otimes k}$. To begin, we recall the Schur-Weyl decomposition of $(\mathbb{C}^n)^{\otimes k}$ as a representation of the group $S_k \times U(n)$.

Proposition 14 (Schur-Weyl duality). *Let k and n be positive integers. Then the decomposition of $(\mathbb{C}^n)^{\otimes k}$ into irreducible subrepresentations of $S_k \times \mathrm{U}(n)$, known as the Schur-Weyl decomposition, is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda' \vdash_n k} W_{\lambda'} \otimes V_{\lambda'}^n \quad \text{\texttt{eq:schur_weyl_decomp}} \quad (34)$$

where the direct summands are indexed by Young diagrams λ' with k boxes and at most n rows, and where $W_{\lambda'}$ supports the irreducible representation of S_k and $V_{\lambda'}^n$ supports the irreducible representation of $\mathrm{U}(n)$.

The dimension of the $V_{\lambda'}^n$ is given by Weyl's dimension formula [Hal08; IN66],

$$\dim(V_{\lambda'}^n) = \frac{\Delta_n(\tilde{\lambda}')}{\Delta_n(\rho)}, \quad \text{\texttt{eq:weyl_dim}} \quad (35)$$

whereas the dimension of $W_{\lambda'}$ may be given by [GW00, Corollary 9.1.5],

$$\dim(W_{\lambda'}) = \frac{|\lambda'|!}{\tilde{\lambda}'_1! \tilde{\lambda}'_2! \cdots \tilde{\lambda}'_n!} \Delta_n(\tilde{\lambda}') \quad \text{\texttt{eq:frob_dim}} \quad (36)$$

where $\tilde{\lambda}' = \lambda' + \rho$, where $\rho = (n-1, n-2, \dots, 1, 0)$ is known as the Weyl vector and where $\Delta_n(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j)$ is the Vandermonde determinant. Note that the representation $W_{\lambda'}$ of S_k is independent of n ; indeed, the right hand side of Eq. (36) does not depend on n and coincides with the Hook-length formula [GW00, Corollary 9.1.6] (any value of n which is at least the depth of λ' suffices).

Definition 15. Given a non-zero positive semidefinite $n \times n$ matrix A , a positive integer x , and a Young diagram $\alpha' \vdash_n x$, let

$$\Gamma[A|\alpha'] := \mathrm{Tr}[\underline{A}^{\otimes a} P_{\alpha'}], \quad (37)$$

where $\underline{A} = \frac{A}{\mathrm{Tr}(A)}$ is the normalization of A and $P_{\alpha'}$ projects onto the isotypic component of $(\mathbb{C}^n)^{\otimes a}$ indexed by α' .

As $P_{\alpha'}$ is $\mathrm{U}(n)$ -equivariant, one observes that $\Gamma[A|\alpha']$ depends only on the eigenvalues of A . The following result, known as spectrum or eigenvalue estimation, states that the map $\alpha' \mapsto \Gamma[A|\alpha']$ for A positive semidefinite is concentrated those Young diagrams α' whose shape matches the shape of the eigenvalues $\alpha = \mathrm{eig}(A)$ [ARS88; KW01; HM02].

\text{\texttt{prop:eig_est}}

Proposition 16 (Eigenvalue estimation). *Let A be a positive semidefinite matrix with eigenvalues $\alpha = \mathrm{eig}(A)$ and let $\alpha' \vdash_n a$. Then*

$$\Gamma[A|\alpha'] \leq \dim(V_{\alpha'}^n) \exp(-aD(\underline{\alpha}'||\underline{\alpha})), \quad (38)$$

where $D(\underline{\alpha}'||\underline{\alpha})$ denotes the Kullback-Leibler divergence of $\underline{\alpha}'$ from $\underline{\alpha}$.

Proof. See [KW01] for a proof (noting that $\mathrm{Tr}(\underline{A}) = 1$). □

\text{\texttt{prop:HS_measure}}

Proposition 17. *Let $d\mu(A)$ denote the $\mathrm{U}(n)$ -invariant Hilbert Schmidt probability measure over the set of $n \times n$ positive semidefinite trace one matrices and let a be a positive integer. Then*

$$\mathbb{E}_A[A^{\otimes a}] := \int d\mu(A) A^{\otimes a} = \sum_{\alpha' \vdash_n a} q_{\alpha'} P_{\alpha'} \quad \text{\texttt{eq:HS_mean}} \quad (39)$$

where the coefficient $q_{\alpha'}$ equals

$$q_{\alpha'} = \frac{\dim(V_{\alpha'}^n)}{\dim(W_{\alpha'}) \dim(V_{(a)}^{n^2})} \quad \text{\texttt{eq:HS_mean_coeff}} \quad (40)$$

Proof. See [ZS01] or [Wal14] for proof details. □

\text{\texttt{prop:HS_integral_uni}}

Proposition 18. *Let A be a positive semidefinite $n \times n$ matrix such that $\mathrm{Tr}(A) = 1$. Then*

$$\mathbb{E}_A \Gamma[A|\alpha'] := \int d\mu(A) \Gamma[A|\alpha'] = \frac{\dim(V_{\alpha'}^n)^2}{\dim(V_{(a)}^{n^2})}. \quad (41)$$

Proposition 19. Let A be positive semidefinite $n \times n$ matrix with $\text{Tr}(A) = 1$ with eigenvalues $\alpha = \text{eig}(A)$ and $\text{rank } r(\alpha) = \text{rank}(A)$. Then for any positive integer a ,

$$\sum_{\alpha' \vdash_{r(\alpha)} a} \Gamma[A|\alpha'] = 1, \quad (42)$$

where the sum is over Young diagrams α' with size a with depth at most $r(\alpha)$.

Definition 20. Given an $n \times n$ matrix X and an Young diagram λ of size $|\lambda|$ define

$$\tau_\lambda(X) := \text{Tr}[P_\lambda X^{\otimes |\lambda|}] \quad (43)$$

Proposition 21. Here we report of a number of properties of $\tau_\lambda(X)$.

- (i) $\tau_{(1)}(X) = \text{Tr}(X)$
- (ii) $\tau_\lambda(I) = \text{Tr}(P_\lambda) = \dim(W_\lambda) \dim(V_\lambda^n)$
- (iii) $\sum_{\lambda \vdash k} \tau_\lambda(X) = \text{Tr}(X)^k$ and thus $\sum_{\lambda \vdash k} \tau_\lambda(\underline{X}) = 1$ for $\underline{X} = \text{Tr}(X)^{-1} X$
- (iv) for all invertible M , $\tau_\lambda(MXM^{-1}) = \tau_\lambda(X)$
- (v) if $X \geq 0$ then $\tau_\lambda(X) \geq 0$
- (vi) if $X, Y \geq 0$, then $\tau_\lambda(X + Y) \geq \tau_\lambda(X) + \tau_\lambda(Y)$
- (vii) if $s \in \mathbb{R}$, then $\tau_\lambda(sX) = s^{|\lambda|} \tau_\lambda(X)$
- (viii) $\tau_\lambda(X)$ is proportional to the Schur-polynomial in the eigenvalues of X :

$$\tau_\lambda(X) = \dim(W_\lambda) s_\lambda(\text{eig}(X)), \quad (44)$$

- (ix) the expected value of $\tau_\lambda(X)$ over all $n \times n$ $X \geq 0$, $\text{Tr}(X) = 1$ with respect to Hilbert-Schmidt measure is

$$\mathbb{E}_X \tau_\lambda(X) = \frac{\dim(V_\lambda^n)^2}{\dim(V_{(|\lambda|)}^{n^2})} \quad (45)$$

- (x) with respect to the Haar measure over $U(n)$,

$$\int_{U(n)} dU (UXU^{-1})^{\otimes k} = \sum_{\lambda \vdash k} \tau_\lambda(X) \frac{P_\lambda}{\text{Tr}(P_\lambda)} \quad (46)$$

Proposition 22. Let $A, B \geq 0$, and suppose $\alpha \vdash a, \beta \vdash b$. Define $C = A + B$ and $c = a + b$. Then

$$\tau_\alpha(A) \tau_\beta(B) \leq \frac{c!}{a!b!} \sum_{\substack{\gamma \vdash c \\ c_{\alpha\beta}^\gamma > 0}} \tau_\gamma(C) \quad (47)$$

Corollary 23.

$$\mathbb{E}_A \tau_\alpha(A) \cdot \mathbb{E}_B \tau_\beta(B) \leq \frac{c!}{a!b!} \sum_{\substack{\gamma \vdash c \\ c_{\alpha\beta}^\gamma > 0}} \mathbb{E}_{A,B} \tau_\gamma(A + B) \quad (48)$$

3.2 Littlewood-Richardson coefficients & trigonal eigenvalues

sec:lr_tri

The purpose of this section is to reveal how eigenvalue estimation can be used to prove a kind of correspondence between trigonal eigenvalue (α, β, γ) and Littlewood-Richardson coefficients $c_{\alpha', \beta'}^{\gamma'}$. Given a triple of Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$, by eigenvalue estimation we see that each of

$$\alpha' \mapsto \Gamma[A|\alpha'], \quad \beta' \mapsto \Gamma[B|\beta'], \quad \gamma' \mapsto \Gamma[C|\gamma'], \quad (49)$$

are respectively concentrated around Young diagrams $(\alpha', \beta', \gamma')$ whose normalized shapes $(\underline{\alpha'}, \underline{\beta'}, \underline{\gamma'})$ approximate the normalized eigenvalues $(\underline{\alpha}, \underline{\beta}, \underline{\gamma})$. As a result of this concentration, one can detect whether there exists Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$ by examining if whether there exists an intersection of subspaces onto which $P_{\alpha'}$, $P_{\beta'}$ and $P_{\gamma'}$ project. To support this claim, we define the following quantity.

Definition 24. Given positive integers a and b , let $c = a + b$. Let $(\alpha', \beta', \gamma')$ be a triple of Young diagrams with sizes:

$$\alpha' \vdash a, \quad \beta' \vdash b, \quad \gamma' \vdash c, \quad (50)$$

{eq:three_yd_sizes}

Define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes a+b}$ by

$$\Pi_{\alpha'} := P_{\alpha'} \otimes \text{id}^{\otimes b}, \quad \Pi_{\beta'} := \text{id}^{\otimes a} \otimes P_{\beta'}, \quad \Pi_{\gamma'} := P_{\gamma'}. \quad (51)$$

{eq:tri_extended_projs}

Given $n \times n$ positive semidefinite matrices A, B and $C = A + B$, let

$$\Gamma[A \ B \ C|\alpha' \ \beta' \ \gamma'] := \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b})(P_{\alpha'} \otimes P_{\beta'})P_{\gamma'}] \quad (52)$$

defn:lr_coeffs

Definition 25 (Littlewood-Richardson coefficients). Let $(\alpha', \beta', \gamma')$ be a triple of Young diagrams with sizes (a, b, c) where $c = a + b$ and let $V_{\alpha'}^n, V_{\beta'}^n, V_{\gamma'}^n$ be a triple of irreducible representations of $U(n)$. Let $C_{\alpha', \beta'}^{\gamma'}$ denote the multiplicity space of $U(n)$ -intertwiners from $V_{\gamma'}$ to $V_{\alpha'} \otimes V_{\beta'}$:

$$C_{\alpha', \beta'}^{\gamma'} := \text{Hom}_{U(n)}(V_{\gamma'}, V_{\alpha'} \otimes V_{\beta'}) \simeq \text{Inv}_{U(n)}(V_{\gamma'^*} \otimes V_{\alpha'} \otimes V_{\beta'}). \quad (53)$$

The *Littlewood-Richardson coefficient* is defined as the dimension of this multiplicity space:

$$c_{\alpha', \beta'}^{\gamma'} := \dim(C_{\alpha', \beta'}^{\gamma'}). \quad (54)$$

The *Clebsch-Gordon isometry* is the $U(n)$ -equivariant isometric linear map

$$Y_{\alpha, \beta}^{\gamma} : C_{\alpha, \beta}^{\gamma} \otimes V_{\gamma} \rightarrow V_{\alpha} \otimes V_{\beta}, \quad (55)$$

sending $f \otimes v \in C_{\alpha', \beta'}^{\gamma'} \otimes V_{\gamma'}$ to $f(v) \in V_{\alpha'} \otimes V_{\beta'}^5$. The Clebsch-Gordon isometries (and coisometries) will be depicted diagrammatically with multiplicity spaces given a distinguished style as in [AB68]:

$$\begin{array}{ccc} \begin{array}{c} V_{\alpha} \quad V_{\beta} \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ C_{\alpha, \beta}^{\gamma} \quad V_{\gamma} \end{array} & \equiv & \begin{array}{c} V_{\alpha} \quad V_{\beta} \\ \diagdown \quad \diagup \\ \boxed{Y_{\alpha, \beta}^{\gamma}} \\ \diagup \quad \diagdown \\ C_{\alpha, \beta}^{\gamma} \quad V_{\gamma} \end{array} \end{array} \quad \begin{array}{ccc} \begin{array}{c} C_{\alpha, \beta}^{\gamma} \quad V_{\gamma} \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ V_{\alpha} \quad V_{\beta} \end{array} & \equiv & \begin{array}{c} C_{\alpha, \beta}^{\gamma} \quad V_{\gamma} \\ \diagdown \quad \diagup \\ \boxed{Y_{\alpha, \beta}^{\gamma, *}} \\ \diagup \quad \diagdown \\ V_{\alpha} \quad V_{\beta} \end{array} \end{array} \quad (56)$$

prop:tri_HS_mean

Proposition 26. Let $(\alpha', \beta', \gamma')$ be Young diagrams with sizes (a, b, c) where $c = a + b$. Then

$$\mathbb{E}_{A, B} \Gamma[A \ B \ C|\alpha' \ \beta' \ \gamma'] = R_{\alpha', \beta'; \gamma'} c_{\alpha', \beta'}^{\gamma'} \quad (57)$$

where $c_{\alpha', \beta'}^{\gamma'}$ is the Littlewood-Richardson coefficient and $R_{\alpha', \beta'; \gamma'}$ is the ratio of dimensions:

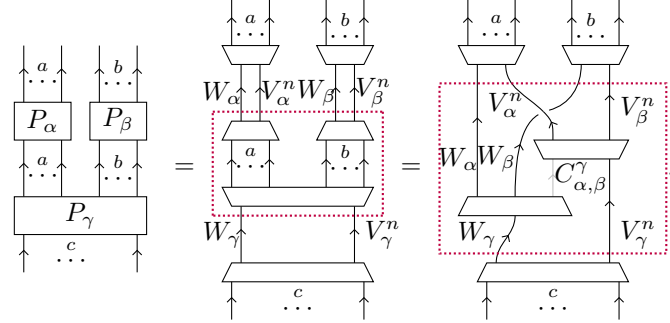
$$R_{\alpha', \beta'; \gamma'} = \frac{\dim(V_{\alpha'}^n) \dim(V_{\beta'}^n) \dim(V_{\gamma'}^n)}{\dim(V_{(a)}^{n^2}) \dim(V_{(b)}^{n^2})}. \quad (58)$$

⁵If $C_{\alpha', \beta'}^{\gamma'} = 0$, then $Y_{\alpha', \beta'}^{\gamma'}$ is identified with the zero map.

Proof. Using Proposition 17 we obtain

$$\mathbb{E}_{A,B} \Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] = q_{\alpha'} q_{\beta'} \text{Tr}[(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'}], \quad \text{\textcolor{blue}{\{eq:mean_tri_gamma\}}} \quad (59)$$

where $q_{\alpha'}$ and $q_{\beta'}$ are given by Eq. (40). Moreover, the operator $(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'}$ decomposes as



The diagram shows the decomposition of the operator $(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'}$ into isometries W and V . The left side shows three boxes $P_{\alpha'}$, $P_{\beta'}$, and $P_{\gamma'}$ stacked vertically. The middle side shows the decomposition into $W_{\alpha'}$, $V_{\alpha'}^n$, $W_{\beta'}$, $V_{\beta'}^n$, and $W_{\gamma'}$. The right side shows the decomposition into $W_{\alpha'}$, $W_{\beta'}$, $C_{\alpha,\beta}^{\gamma'}$, and $W_{\gamma'}$. The diagram is labeled with \text{\textcolor{blue}{\{eq:lr_triple_prod\}}} and (60).

and therefore taking the trace yields

$$\text{Tr}[(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'}] = \dim(W_{\alpha'}) \dim(W_{\beta'}) \dim(C_{\alpha,\beta}^{\gamma'}) \dim(V_{\gamma'}^n). \quad \text{\textcolor{blue}{\{eq:trace_tri_formula\}}} \quad (61)$$

Combining Eq. (59) and Eq. (61) produces the claimed formula as $c_{\alpha',\beta'}^{\gamma'} = \dim(C_{\alpha,\beta}^{\gamma'})$. $\square$

Our first result establishes a connection between $\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']$ and the corresponding Littlewood-Richardson coefficient.

$\text{\textcolor{blue}{cor:tri_indep_upper_bound}}$

Corollary 27. *Let $(\alpha', \beta', \gamma')$ be Young diagrams with sizes (a, b, c) satisfying $c = a + b$. Then for any positive semidefinite matrices (A, B, C) satisfying $C = A + B$, we have*

$$c_{\alpha',\beta'}^{\gamma'} = 0 \implies \Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] = 0. \quad \text{\textcolor{blue}{\{eq:tri_gamma_lr_upper_bound\}}} \quad (62)$$

Proof. From the definition of $\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']$, and the fact that $\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b}$ is trace one positive semidefinite, we have the bound

$$\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] \leq \|(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'}\|_{\infty} \quad (63)$$

Using the decomposition of $(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'}$ into isometries given by Eq. (60), we also have

$$\|(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'}\|_{\infty} = \left\| \text{id}_{C_{\alpha,\beta}^{\gamma'}} \right\|_{\infty} = \begin{cases} 1 & c_{\alpha',\beta'}^{\gamma'} > 0 \\ 0 & c_{\alpha',\beta'}^{\gamma'} = 0 \end{cases}. \quad (64)$$

Therefore if $c_{\alpha',\beta'}^{\gamma'} = 0$, then $(P_{\alpha'} \otimes P_{\beta'}) P_{\gamma'} = 0$ and thus $\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] = 0$ also. $\square$

The following proposition is the analogue of Proposition 19.

$\text{\textcolor{blue}{prop:tri_completeness}}$

Proposition 28. *Let (A, B, C) be positive semidefinite $n \times n$ matrices satisfying $C = A + B$ with eigenvalues (α, β, γ) . Then*

$$\sum_{\alpha', \beta', \gamma'} \Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] = 1. \quad \text{\textcolor{blue}{\{eq:tri_completeness\}}} \quad (65)$$

where the sum is taken over Young diagrams with the following sizes:

$$\alpha' \vdash_{r(\alpha)} a, \quad \beta' \vdash_{r(\beta)} b, \quad \gamma' \vdash_{r(\gamma)} c. \quad \text{\textcolor{blue}{\{eq:three_yd_sizes_rank\}}} \quad (66)$$

Proof. To begin, note that Eq. (65) holds when the sum is taken over Young diagrams with unrestricted rank, meaning $(\alpha', \beta', \gamma')$ have sizes (a, b, c) with $c = a + b$ but arbitrary rank/depth. That one can restrict to Young diagrams with depths bounded according to Eq. (66) follows because $\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] = 0$ whenever the depth of *any* of $(\alpha', \beta', \gamma')$ exceeds the corresponding ranks $(r(\alpha), r(\beta), r(\gamma))$. $\square$

Theorem 29. Let (α, β, γ) be a triple of eigenvalues of non-zero positive semidefinite (A, B, C) satisfying $C = A + B$. Let $(\alpha', \beta', \gamma')$ be Young diagrams with sizes given by

$$\alpha' \vdash a = \lceil k |\alpha| \rceil, \quad \beta' \vdash b = \lceil k |\beta| \rceil, \quad \gamma' \vdash c = \lceil k |\alpha| \rceil + \lceil k |\beta| \rceil, \quad \text{\textcolor{blue}{\eq:ceil_tri_yd_sizes}} \quad (67)$$

for any positive integer $k \geq \tau_{\min}^{-1}$ where $\tau_{\min} = \min\{|\alpha|, |\beta|, |\gamma|\}$. Then

$$\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] \leq p_{\Delta}(k) \exp(-k I_{\alpha\beta\gamma}^{\alpha'\beta'\gamma'} / 2). \quad \text{\textcolor{blue}{\eq:tri_gamma_upper_bound}} \quad (68)$$

where $p_{\Delta}(k)$ grows polynomially in k and where $I_{\alpha\beta\gamma}^{\alpha'\beta'\gamma'}$ is a quantity which measures the deviation of the primed quantities from the unprimed quantities:

$$I_{\alpha\beta\gamma}^{\alpha'\beta'\gamma'} = |\alpha| D(\underline{\alpha}' \| \underline{\alpha}) + |\beta| D(\underline{\beta}' \| \underline{\beta}) + |\gamma| D(\underline{\gamma}' \| \underline{\gamma}). \quad (69)$$

Proof. The first portion of the proof does not depend on the value of k . For now, we suppose the Young diagrams $(\alpha', \beta', \gamma')$ have sizes (a, b, c) with $c = a + b$. Then

$$\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']^2 = \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b})(P_{\alpha'} \otimes P_{\beta'})P_{\gamma'}]^2, \quad (70)$$

$$\leq \text{Tr}[\underline{A}^{\otimes a} P_{\alpha'}] \text{Tr}[\underline{B}^{\otimes b} P_{\beta'}] \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b})P_{\gamma'}] \quad (71)$$

The third term can be relaxed further to an upper bound in terms of $C = A + B$ because $C^{\otimes c} = (A + B)^{\otimes a+b}$ and therefore

$$\text{Tr}(P_{\gamma'} C^{\otimes c}) = \sum_{j=0}^c \binom{c}{j} \text{Tr}(P_{\gamma'} (A^{\otimes j} \otimes B^{\otimes c-j})). \quad (72)$$

Here we have used the S_c -equivariance of $P_{\gamma'}$. Since each of the summands above are non-negative, we conclude for all $0 \leq j \leq c$ that

$$\text{Tr}(P_{\gamma'} C^{\otimes c}) \geq \binom{c}{j} \text{Tr}(P_{\gamma'} (A^{\otimes j} \otimes B^{\otimes c-j})). \quad (73)$$

Letting $j = a$ and $c - j = c - a = b$ and rearranging yields

$$\text{Tr}(P_{\gamma'} (A^{\otimes a} \otimes B^{\otimes b})) \leq \frac{a!b!}{c!} \text{Tr}(P_{\gamma'} C^{\otimes c}). \quad (74)$$

Altogether we obtain

$$\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']^2 \leq \frac{a!b!}{c!} \frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \text{Tr}[\underline{A}^{\otimes a} P_{\alpha'}] \text{Tr}[\underline{B}^{\otimes b} P_{\beta'}] \text{Tr}[\underline{C}^{\otimes c} P_{\gamma'}] \quad \text{\textcolor{blue}{\eq:tri_gamma_abc_upper_bound}} \quad (75)$$

Our objective now will be to use the k -dependent values for (a, b, c) to relax this upper bound further. From Eq. (67), we obtain the following estimates:

$$k |\alpha| \leq a < k |\alpha| + 1, \quad k |\beta| \leq b < k |\beta| + 1, \quad k |\gamma| \leq c < k |\gamma| + 2. \quad (76)$$

An application of Lemma 48 for binomial distributions ($m = 2$) implies the following bound

$$\frac{c!}{a!b!} \frac{|\alpha|^a |\beta|^b}{|\gamma|^c} \geq 12^2 (k |\gamma| + 2)^{\frac{1}{2}}. \quad (77)$$

Moreover, from Weyl's dimension formula, Eq. (35), we obtain [CM06, Equation 7]:

$$\dim(V_{\gamma'}^n) \leq (|\gamma'| + 1)^{n(n-1)/2} \leq (k |\gamma| + 3)^{n(n-1)/2}, \quad (78)$$

because $(\gamma'_i - i) - (\gamma'_j - j) \leq |\gamma'| + 1$ as well as $j - i \geq 1$. Using eigenvalue estimation, Proposition 16, we can relax the bound on $\text{Tr}[\underline{C}^{\otimes c} P_{\gamma'}]$ to obtain a bound in terms of k :

$$\text{Tr}[\underline{C}^{\otimes c} P_{\gamma'}] \leq \dim(V_{\gamma'}^n) \exp(-c D(\underline{\gamma}' \| \underline{\gamma})) \leq (k |\gamma| + 3)^{n(n-1)/2} \exp(-k |\gamma| D(\underline{\gamma}' \| \underline{\gamma})) \quad (79)$$

with similar bounds on $\text{Tr}[A^{\otimes a} P_{\alpha'}]$ and $\text{Tr}[B^{\otimes b} P_{\beta'}]$. Combining these bounds together yields

$$\Gamma[A \ B \ C | \alpha'_k \ \beta'_k \ \gamma'_k]^2 \leq 12^2 (k |\gamma| + 2)^{\frac{1}{2}} [(k |\alpha| + 2)(k |\beta| + 2)(k |\gamma| + 3)]^{n(n-1)/2} \exp(-k I_{\alpha\beta\gamma}^{\alpha' \beta' \gamma'}). \quad (80)$$

Therefore, the claim holds if $p_{\Delta}(k)$ is taken to be

$$p_{\Delta}(k) = 12(k\tau_{\max} + 3)^{\frac{3n(n-1)+1}{4}}. \quad \text{\textcolor{blue}{\eq:p_of_k_tri}} \quad (81)$$

where $\tau_{\max} = \max\{|\alpha|, |\beta|, |\gamma|\}$. $\square$

While the previous theorem states that $\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']$ concentrates in magnitude around particular Young diagrams, it is not directly applicable to Horn problem simply because its value depends on more than just the eigenvalues of the matrices (A, B, C) . Nevertheless, combining the previous theorem the completeness relation (Eq. (65)) yields a necessary condition for (α, β, γ) to be tetrahedral.

Corollary 30. *Let (α, β, γ) be the eigenvalues of non-zero positive semidefinite matrices (A, B, C) satisfying $C = A+B$ with minimum trace $\tau_{\min}$. Then there exists a sextuple of sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k)$ indexed by positive integers $k > \tau_{\min}^{-1}$ with sizes given by Eq. (67) satisfying (i) the convergence condition*

$$I_{\alpha\beta\gamma}^{\alpha'_k \beta'_k \gamma'_k} \leq \frac{2}{k} \log p_{\Delta}(k) q_{\Delta}(k) \xrightarrow{k \rightarrow \infty} 0, \quad \text{\textcolor{blue}{\eq:tri_convergence_rate}} \quad (82)$$

where $q_{\Delta}(k) = (k\tau_{\max} + 2)^{3(r_{\max}-1)}$, and (ii) the non-vanishing of the Littlewood-Richardson coefficient

$$c_{\alpha'_k, \beta'_k}^{\gamma'_k} \neq 0. \quad (83)$$

Proof. To proof the claim, we construct the sequence implicitly. For each k , it suffices to take $(\alpha'_k, \beta'_k, \gamma'_k)$ to be any sextuple of Young diagrams which maximizes the value of $\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']$ with sizes determined by Eq. (67):

$$(\alpha'_k, \beta'_k, \gamma'_k) := \underset{\alpha' \beta' \gamma'}{\operatorname{argmax}} \Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']. \quad (84)$$

While this maximal choice may not be unique, it will necessarily satisfy a lower bound due to the completeness relation satisfied by $\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma']$ Eq. (65). Specifically, this maximal choice must satisfy

$$\Gamma[A \ B \ C | \alpha'_k \ \beta'_k \ \gamma'_k] \geq \frac{1}{q_{\Delta}(k)} \quad \text{\textcolor{blue}{\eq:tri_gamma_lower_bound}} \quad (85)$$

where $q_{\Delta}(k)$ is any upper bound on the number of possible summands in the completeness relation Eq. (65) viewed as a function of k . As there less than or equal to $(s+1)^{r-1}$ Young diagrams with r rows with size at most s , letting $q_{\Delta}(k)$ be as stated in the claim ensures Eq. (85) holds because the size (resp. rank) of any of $(\alpha'_k, \beta'_k, \gamma'_k)$ is at most $k\tau_{\max} + 2$ (resp. $r_{\max}$). Moreover, applying Eq. (68) yields

$$I_{\alpha\beta\gamma}^{\alpha'_k \beta'_k \gamma'_k} \leq \frac{2}{k} \log(q_{\Delta}(k) p_{\Delta}(k)) \quad (86)$$

and thus Eq. (82) holds because $p_{\Delta}(k)$ and $q_{\Delta}(k)$ are polynomials in k with coefficients depending on the maximum trace $\tau_{\max}$ and maximum rank $r_{\max}$.

The non-vanishing of the Littlewood-Richardson coefficient is implied by Corollary 27. $\square$

Remark 31. The convergence condition above (Eq. (82)) in conjunction with the size conditions given by Eq. (67) together ensure that the Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k)$ converge to the eigenvalues (α, β, γ) in the following sense that

$$\frac{\alpha'_k}{k} \rightarrow \alpha, \quad \frac{\beta'_k}{k} \rightarrow \beta, \quad \frac{\gamma'_k}{k} \rightarrow \gamma, \quad \text{\textcolor{blue}{\eq:tri_convergence}} \quad (87)$$

where $\frac{\lambda'_k}{k} \rightarrow \lambda$ is shorthand for $\lim_{k \rightarrow \infty} (\lambda'_k)_i / k = \lambda_i$ for all $i \in \{1, \dots, n\}$.

While the existence of a convergent sequence $(\alpha'_k, \beta'_k, \gamma'_k)$ of primed Young diagrams is necessary for (α, β, γ) to be tetrahedral by Corollary 41, our next result proves it is also sufficient. In other words, if (α, β, γ) are not tetrahedral, then no such sequence can exist.

Corollary 32. *Let (α, β, γ) be a non-trigonal triple of positive semidefinite and non-zero eigenvalues. Then for all sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k)$ with sizes given by Eq. (67) and converging to (α, β, γ) in the sense that Eq. (82) holds, one necessarily has*

$$\forall k \geq K : c_{\alpha'_k, \beta'_k}^{\gamma'_k} = 0. \quad (88)$$

for some sufficiently large K .

Proof. **TODO:** write the proof using Proposition 26. □

TODO: Summarize [Chr08]. **TODO:** Briefly mention the direct and pedagogical Kempf-Ness style proofs of stronger results in [BVW18, Proposition 2.3] also using the strong duality result of [FW20, Theorem 1 & Proposition 3.1] and the quantitative Kempf-Ness theorem [Bür+19, Theorem 1.17].

3.3 Generalized $6j$ symbols & tetrahedral eigenvalues

sec:sixj-tet

The purpose of this section is to introduce a representation theoretic object, namely the generalized Wigner $6j$ symbol, and then show how its asymptotic properties encode whether a given sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is tetrahedral in a manner which analogous to how the Littlewood-Richardson coefficient encodes whether a given triple (α, β, γ) is trigonal. For the case where $n = 2$, this correspondence is well-known [Rob99; PR68; Bac10], while for the case where $n \geq 3$, this correspondence was already established indirectly by Christandl, Şahinoğlu, and Walter [CŞW18]. Here we establish this correspondence directly.

defn:tet_gamma

Definition 33. Given positive integers a, b , and d , let $c = a + b$, $f = b + d$, and $e = a + b + d$. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of Young diagrams with sizes:

$$\alpha' \vdash a, \quad \beta' \vdash b, \quad \gamma' \vdash c, \quad \delta' \vdash d, \quad \epsilon' \vdash e, \quad \phi' \vdash f. \quad (89)$$

{eq:six_yd_sizes}

Define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes a+b+d}$ by

$$\begin{aligned} \Pi_{\alpha'} &:= P_{\alpha'} \otimes \text{id}^{\otimes b} \otimes \text{id}^{\otimes d}, & \Pi_{\beta'} &:= \text{id}^{\otimes a} \otimes P_{\beta'} \otimes \text{id}^{\otimes d}, & \Pi_{\gamma'} &:= P_{\gamma'} \otimes \text{id}^{\otimes d}, \\ \Pi_{\delta'} &:= \text{id}^{\otimes a} \otimes \text{id}^{\otimes b} \otimes P_{\delta'}, & \Pi_{\epsilon'} &:= P_{\epsilon'}, & \Pi_{\phi'} &:= \text{id}^{\otimes a} \otimes P_{\phi'}. \end{aligned} \quad (90)$$

{eq:tet_extended_projs}

Given $n \times n$ positive semidefinite matrices A, B, D with $C = A + B$, $F = B + D$, and $E = A + B + D$, let

$$\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} = \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}]. \quad (91)$$

Remark 34. In the analysis that follows, it will be convenient to avoid certain edge cases pertaining to the candidate sextuple of positive semidefinite eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Specifically, let the extremal traces be denoted by

$$\tau_{\min} := \min\{|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|\}, \quad \tau_{\max} := \max\{|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|\}, \quad (92)$$

and the extremal ranks be denoted by

$$r_{\min} := \min\{r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi)\}, \quad r_{\max} := \max\{r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi)\}. \quad (93)$$

In order to avoid the edge case where one of the eigenvalues $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$ is totally zero, we sometimes demand that the minimum trace is strictly positive or equivalently that the minimum rank is at least one:

$$\tau_{\min} > 0, \quad r_{\min} \geq 1. \quad (94)$$

Another technical difficulty which complicates subsequent analysis is that the set of projection operators $\{\Pi_{\alpha'}, \Pi_{\beta'}, \Pi_{\gamma'}, \Pi_{\delta'}, \Pi_{\epsilon'}, \Pi_{\phi'}\}$ defined by Eq. (90) is *almost* pairwise mutually commuting.

Lemma 35. Every pair of projection operators in $\{\Pi_{\alpha'}, \Pi_{\beta'}, \Pi_{\gamma'}, \Pi_{\delta'}, \Pi_{\epsilon'}, \Pi_{\phi'}\}$ is commuting with only possible exception being $\Pi_{\gamma'}$ and $\Pi_{\phi'}$. Therefore, the operators $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ and $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ defined by

$$\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'} = \Pi_{\gamma'} \Pi_{\alpha'} \Pi_{\beta'} \Pi_{\delta'} \Pi_{\epsilon'}, \quad \text{and} \quad \Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} = \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Pi_{\delta'} \Pi_{\epsilon'}, \quad (95)$$

are projection operators.

Proof. That all projection operators except the pair $\{\Pi_{\gamma'}, \Pi_{\phi'}\}$ commute follows from elementary applications of Schur's lemma. The obstruction for the pair $\{\Pi_{\gamma'}, \Pi_{\phi'}\}$ is simply that $\Pi_{\gamma'}$ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes b+d}$ and likewise $\Pi_{\phi'}$ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes a+b}$. $\square$

defn:sixj

Definition 36 (Generalized 6j operators). Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of Young diagrams with sizes (a, b, c, d, e, f) given by Eq. (89). Then we define the $\text{U}(n)$ -6j operator to be the operator⁶

$$\left\{ \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right\} : C_{\alpha', \beta'}^{\gamma'} \otimes C_{\delta', \gamma'}^{\epsilon'} \rightarrow C_{\alpha', \phi'}^{\epsilon'} \otimes C_{\beta', \delta'}^{\phi'}, \quad (96)$$

defined by

$$\left\{ \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right\} := \frac{1}{\dim(V_{\epsilon'})} \text{Tr}_{V_{\epsilon'}} [Y_{\alpha', \phi'}^{\epsilon' \dagger} (\text{id}_{V_{\alpha'}} \otimes Y_{\beta', \delta'}^{\phi' \dagger}) (Y_{\alpha', \beta'}^{\gamma'} \otimes \text{id}_{V_{\delta'}}) Y_{\gamma', \delta'}^{\epsilon'}], \quad (97)$$

eq:sixj

and depicted diagrammatically by

The following proposition is the analogue of Proposition 18.

prop:HS_integral_tet

Proposition 37. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams with sizes given by Eq. (89). Then

$$\begin{aligned} \mathbb{E}_{A,B,D} \Gamma \left[\begin{matrix} A & B & C \\ D & E & F \end{matrix} \middle| \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right] &:= \iiint d\mu(A) d\mu(B) d\mu(D) \Gamma \left[\begin{matrix} A & B & C \\ D & E & F \end{matrix} \middle| \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right] \\ &= R_{\alpha'\beta'\delta';\epsilon'} \left\| \left\{ \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right\} \right\|_{\text{HS}}^2 \end{aligned} \quad (99)$$

eq:tet_mean_value

where $R_{\alpha'\beta'\delta';\epsilon'}$ is a ratio of dimensions:

$$R_{\alpha'\beta'\delta';\epsilon'} = \frac{\dim(V_{\alpha'}^n) \dim(V_{\beta'}^n) \dim(V_{\delta'}^n) \dim(V_{\epsilon'}^n)}{\dim(V_{(a)}^{n^2}) \dim(V_{(b)}^{n^2}) \dim(V_{(d)}^{n^2})}. \quad (100)$$

⁶It is important to note that there does not seem to exist a universal convention concerning how 6j symbols are normalized; Coquereaux discusses this problem in [Coq23].

Proof. To prove Eq. (99), we use Proposition 17 and Lemma 35 to see that

$$\mathbb{E}_{A,B,D}\Gamma\left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}\right] = q_{\alpha'} q_{\beta'} q_{\delta'} \text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}). \quad \text{\textcolor{blue}{\code{eq:tet_mean_step_1}}} \quad (101)$$

We claim that the product $\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}$ is related to the generalized $6j$ operator $\left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\}$ in that there exists appropriately-typed isometries ι_I and ι_{II} such that

$$\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'} = \iota_{II} \left(\text{id}_{V_{\epsilon}^n} \otimes \text{id}_{W_{\alpha}} \otimes \text{id}_{W_{\beta}} \otimes \text{id}_{W_{\delta}} \otimes \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right) \iota_I^\dagger \quad \text{\textcolor{blue}{\code{eq:ud_sixj_prod_proj_iso}}} \quad (102)$$

The form of these isometries can be deduced by applying Eq. (60) four times in the following diagrammatic calculation:

From Eq. (102) we conclude

$$\text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}) = \dim(V_{\epsilon}^n) \dim(W_{\alpha}) \dim(W_{\beta}) \dim(W_{\delta}) \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}, \quad \text{\textcolor{blue}{\code{eq:ud_sixj_prod_proj_hs_norm}}} \quad (104)$$

Combining Eq. (104) with Eq. (101) and Eq. (40) then yields Eq. (99). $\square$

The following proposition is the analogue of Proposition 19.

$\text{\textcolor{blue}{\code{prop:completeness_rel_tet}}}$

Proposition 38. Let (A, B, C, D, E, F) be tetrahedral positive semidefinite matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Then

$$\sum_{\substack{\alpha' \vdash_{r(\alpha)} a \\ \beta' \vdash_{r(\beta)} b \\ \gamma' \vdash_{r(\gamma)} c \\ \delta' \vdash_{r(\delta)} d \\ \epsilon' \vdash_{r(\epsilon)} e \\ \phi' \vdash_{r(\phi)} f}} \Gamma\left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}\right] = 1. \quad \text{\textcolor{blue}{\code{eq:tet_completeness}}} \quad (105)$$

where the sum is taken over Young diagrams with the following sizes:

$$\alpha' \vdash_{r(\alpha)} a, \quad \beta' \vdash_{r(\beta)} b, \quad \gamma' \vdash_{r(\gamma)} c, \quad \delta' \vdash_{r(\delta)} d, \quad \epsilon' \vdash_{r(\epsilon)} e, \quad \phi' \vdash_{r(\phi)} f. \quad \text{\textcolor{blue}{\code{eq:six_yd_sizes_rank}}} \quad (106)$$

Proof. To begin, note that Eq. (105) holds when the sum is taken over Young diagrams with unrestricted rank according to Eq. (89) because $\sum_{\lambda' \vdash n} P_{\lambda'} = \text{id}_n^{\otimes \ell}$ for all positive integers ℓ and $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$. That one can restrict to Young diagrams with depths bounded according to Eq. (106) follows because $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} = 0$ whenever the depth of *any* of $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ exceeds the corresponding ranks $(r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi))$.⁷ $\square$

Our next goal will be to derive two upper bounds on the absolute value of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$. Our first upper bound is *independent* of the given matrices and follows as a corollary of Lemma 35. cor:abd_indep_upper_bound

Corollary 39. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams of sizes given by Eq. (89). Then for all tetrahedral, non-zero positive semidefinite (A, B, C, D, E, F) , we have*

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| \leq \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\infty}. \quad \text{feq:tet_gamma_size_upper_bound} \quad (107)$$

Proof. From Eq. (95) we have

$$\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} = \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}). \quad (108)$$

Moreover, as $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$ is positive semidefinite trace one, and $\Pi_1 := \Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ and $\Pi_2 := \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ are projection operators by Lemma 35, we have

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| = |\text{Tr}(\Pi_1 \Omega \Pi_2)| \leq |\langle v, \Pi_1 \Pi_2 v \rangle| \leq \|\Pi_1 \Pi_2\|_{\infty} = \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\infty}. \quad (109)$$

where the last equality follows from the isometry decomposition provided by Eq. (102). $\square$

Our second upper bound on the magnitude of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ depends on the *eigenvalues* $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of the given tetrahedral matrices (A, B, C, D, E, F) where $C = A + B$, $F = B + D$ and $E = A + B + D$. This result proves that in the limit of large Young diagrams, the magnitude of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ is sharply concentrated around those Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ whose shapes resemble the sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of (A, B, C, D, E, F) . thm:abd_dep_upper_bound

Theorem 40. *Let everything be as in Definition 33 and let the Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ have sizes given by*

$$\begin{aligned} \alpha' \vdash a &= \lceil k |\alpha| \rceil, & \beta' \vdash b &= \lceil k |\beta| \rceil, & \gamma' \vdash c &= \lceil k |\alpha| \rceil + \lceil k |\beta| \rceil, & \text{feq:ceil_yd_sizes} \\ \delta' \vdash d &= \lceil k |\delta| \rceil, & \epsilon' \vdash e &= \lceil k |\alpha| \rceil + \lceil k |\beta| \rceil + \lceil k |\delta| \rceil, & \phi' \vdash f &= \lceil k |\beta| \rceil + \lceil k |\delta| \rceil, \end{aligned} \quad (110)$$

for some positive integer $k > \tau_{\min}^{-1}$ where $\tau_{\min}$ is the minimum trace. Then

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| \leq p(k) \exp(-k I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} / 6). \quad \text{feq:tet_gamma_upper_bound} \quad (111)$$

where $p(k)$ grows polynomially in k and where $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$ is a quantity which measures the deviation of the primed quantities from the unprimed quantities:

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = |\alpha| D(\underline{\alpha'} \parallel \underline{\alpha}) + |\beta| D(\underline{\beta'} \parallel \underline{\beta}) + |\gamma| D(\underline{\gamma'} \parallel \underline{\gamma}) + |\delta| D(\underline{\delta'} \parallel \underline{\delta}) + |\epsilon| D(\underline{\epsilon'} \parallel \underline{\epsilon}) + |\phi| D(\underline{\phi'} \parallel \underline{\phi}). \quad (112)$$

Proof. The first portion of the proof does not depend on the value of k . For now, we suppose the Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ have sizes determined by Eq. (89). As in the proof of Corollary 39, we let $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$. Using the commutation relations provided by Lemma 35 we apply the Cauchy-Schwarz

⁷This claim actually follows from Theorem 40 and will be proven shortly. Importantly, Theorem 40 does not rely on Proposition 38.

inequality to $\langle X, Y \rangle = \text{Tr}(X^*Y) \leq \text{Tr}(X^*X)^{\frac{1}{2}} \text{Tr}(Y^*Y)^{\frac{1}{2}}$ for $X = \Omega^{\frac{1}{2}} \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}$ and $Y = \Omega^{\frac{1}{2}} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'}$ to obtain

$$\begin{aligned} \left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} \right|^6 &= |\text{Tr}[(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}) \Omega (\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'})]|^6 \\ &\leq \text{Tr}(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Omega)^3 \text{Tr}(\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Omega)^3 \\ &\leq \text{Tr}(\Pi_{\phi'} \Omega) \text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega), \end{aligned} \quad (113)$$

where the last inequality follows because $\Pi_{\lambda'} \leq \text{id}_n^{\otimes |\lambda'|}$ for each λ' in $\{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$.

Now, for each $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$, it is possible to upper-bound $\text{Tr}(\Pi_{\lambda'} \Omega)$ using the eigenvalue estimation technique (Proposition 16). For a detailed example, we will consider value of $\text{Tr}(\Pi_{\gamma'} \Omega)$:

$$\text{Tr}(\Pi_{\gamma'} \Omega) = \frac{\text{Tr}[(P_{\gamma'} \otimes \text{id}^{\otimes d})(A^{\otimes a} \otimes B^{\otimes b} \otimes D^{\otimes d})]}{|\alpha|^a |\beta|^b |\delta|^d} = \frac{\text{Tr}[P_{\gamma'}(A^{\otimes a} \otimes B^{\otimes b})]}{|\alpha|^a |\beta|^b}. \quad (114)$$

Under the assumption that $C = A + B$, we have $C^{\otimes c} = (A + B)^{\otimes a+b}$ which implies

$$\text{Tr}(P_{\gamma'} C^{\otimes c}) = \sum_{k=0}^c \binom{c}{k} \text{Tr}(P_{\gamma'}(A^{\otimes k} \otimes B^{\otimes c-k})). \quad (115)$$

Here we have used the S_c -equivariance of $P_{\gamma'}$. Since each of the summands above are non-negative, we conclude for all $0 \leq j \leq c$ that

$$\text{Tr}(P_{\gamma'} C^{\otimes c}) \geq \binom{c}{j} \text{Tr}(P_{\gamma'}(A^{\otimes j} \otimes B^{\otimes c-j})). \quad (116)$$

Letting $j = a$ and rearranging yields

$$\text{Tr}(P_{\gamma'}(A^{\otimes a} \otimes B^{\otimes b})) \leq \frac{a!b!}{c!} \text{Tr}(P_{\gamma'} C^{\otimes c}). \quad (117)$$

Combining this result with the spectrum estimation bound (Proposition 16) yields

$$\text{Tr}(\Pi_{\gamma'} \Omega) \leq \frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \dim(V_{\gamma'}^n) \exp(-cD(\underline{\gamma}' \parallel \underline{\gamma})). \quad (118)$$

Analogous calculations for $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$ produce the following set of upper bounds:

$$\begin{aligned} \text{Tr}(\Pi_{\alpha'} \Omega) &\leq \dim(V_{\alpha'}^n) \exp(-aD(\underline{\alpha}' \parallel \underline{\alpha})), \\ \text{Tr}(\Pi_{\beta'} \Omega) &\leq \dim(V_{\beta'}^n) \exp(-bD(\underline{\beta}' \parallel \underline{\beta})), \\ \text{Tr}(\Pi_{\gamma'} \Omega) &\leq \dim(V_{\gamma'}^n) \frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \exp(-cD(\underline{\gamma}' \parallel \underline{\gamma})), \\ \text{Tr}(\Pi_{\delta'} \Omega) &\leq \dim(V_{\delta'}^n) \exp(-dD(\underline{\delta}' \parallel \underline{\delta})), \\ \text{Tr}(\Pi_{\epsilon'} \Omega) &\leq \dim(V_{\epsilon'}^n) \frac{|\epsilon|^e}{|\alpha|^a |\beta|^b |\delta|^d} \frac{a!b!d!}{e!} \exp(-eD(\underline{\epsilon}' \parallel \underline{\epsilon})), \\ \text{Tr}(\Pi_{\phi'} \Omega) &\leq \dim(V_{\phi'}^n) \frac{|\phi|^f}{|\beta|^b |\delta|^d} \frac{b!d!}{f!} \exp(-fD(\underline{\phi}' \parallel \underline{\phi})). \end{aligned} \quad (119)$$

Our objective now will be to use the k -dependent values for (a, b, c, d, e, f) to upper bound the multinomial prefactors appearing in the bounds for γ' , ϵ' , and ϕ' above in terms of k . From Eq. (110), we obtain the following estimates:

$$\begin{aligned} k|\alpha| &\leq a < k|\alpha| + 1, & k|\beta| &\leq b < k|\beta| + 1, & k|\gamma| &\leq c < k|\gamma| + 2, & \text{\textcolor{blue}{\texttt{eq:size_estimates}}} \\ k|\delta| &\leq d < k|\delta| + 1, & k|\epsilon| &\leq e < k|\epsilon| + 3, & k|\phi| &\leq f < k|\phi| + 2. \end{aligned} \quad (120)$$

An application of Lemma 48 for binomial distributions ($m = 2$) implies the following bound

$$\frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \leq 12^2 (k|\gamma| + 2)^{\frac{1}{2}}. \quad (121)$$

Moreover, from Weyl's dimension formula, Eq. (35), we obtain [CM06, Equation 7]:

$$\dim(V_{\gamma'}^n) \leq (|\gamma'| + 1)^{n(n-1)/2} \leq (k|\gamma| + 3)^{n(n-1)/2}, \quad (122)$$

because $(\gamma'_i - i) - (\gamma'_j - j) \leq |\gamma'| + 1$ as well as $j - i \geq 1$. Altogether, we can relax the bound on $\text{Tr}(\Pi_{\gamma'}\Omega)$ to obtain

$$\text{Tr}(\Pi_{\gamma'}\Omega) \leq 12^2 (k|\gamma| + 3)^{\frac{n(n-1)}{2} + \frac{1}{2}} \exp(-k|\gamma| D(\underline{\gamma}' \parallel \underline{\gamma})). \quad (123)$$

A similar calculations for $\text{Tr}(\Pi_{\epsilon'}\Omega)$ yields (noting to apply Lemma 48 for trinomial distributions when $m = 3$) yields:

$$\text{Tr}(\Pi_{\epsilon'}\Omega) \leq 12^3 (k|\epsilon| + 4)^{\frac{n(n-1)}{2} + 1} \exp(-k|\epsilon| D(\underline{\epsilon}' \parallel \underline{\epsilon})). \quad (124)$$

It is also not too difficult to see that the polynomial in k prefactor in the upper bound on $\text{Tr}(\Pi_{\epsilon'}\Omega)$ above is the largest such prefactor and thus Eq. (111) indeed holds if we take $p(k)$ to be

$$p(k) = 12^3 (k\tau_{\max} + 4)^{\frac{n(n-1)}{2} + 1}. \quad \text{\textcolor{blue}{\code{eq:p_of_k}}} \quad (125)$$

□

While Theorem 40 states that $\Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right]$ concentrates in magnitude around particular Young diagrams, it is not directly applicable to the tetrahedral Horn problem simply because its value depends on more than just the eigenvalues of the matrices (A, B, C, D, E, F) . Nevertheless, combining Theorem 40 with the completeness relation (Eq. (105)) yields a necessary condition for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral. \text{\textcolor{blue}{\code{cor:tet_size_lower_bound}}}

Corollary 41. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of non-zero positive semidefinite tetrahedral Hermitian matrices (A, B, C, D, E, F) with minimum trace $\tau_{\min}$. Then there exists a sextuple of sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ indexed by positive integers $k > \tau_{\min}^{-1}$ with sizes given by Eq. (110) satisfying both (i) the convergence condition*

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k} \leq \frac{6}{k} \log p(k)q(k) \xrightarrow{k \rightarrow \infty} 0, \quad \text{\textcolor{blue}{\code{eq:sextuple_convergence_rate}}} \quad (126)$$

and (ii) the lower bound

$$\left\| \left\{ \begin{smallmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{smallmatrix} \right\} \right\|_{\infty} \geq \frac{1}{q(k)}, \quad \text{\textcolor{blue}{\code{eq:ud_size_lower_bound}}} \quad (127)$$

where $p(k)$ is given by Eq. (125) whereas $q(k)$ is given by

$$q(k) = (k\tau_{\max} + 4)^{6(r_{\max}-1)}. \quad \text{\textcolor{blue}{\code{eq:q_of_k}}} \quad (128)$$

Proof. To proof the claim, we construct the sequence implicitly. For each k , it suffices to take $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ to be any sextuple of Young diagrams which maximizes the value of $\left| \Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \right|$ with sizes determined by Eq. (110):

$$(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k) := \underset{\alpha' \beta' \gamma' \delta' \epsilon' \phi'}{\operatorname{argmax}} \left| \Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \right|. \quad (129)$$

While this maximal choice may not be unique, it will necessarily satisfy a lower bound due to the completeness relation satisfied by $\Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right]$ Eq. (105). Specifically, this maximal choice must satisfy

$$\left| \Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{smallmatrix} \right] \right| \geq \frac{1}{q(k)} \quad \text{\textcolor{blue}{\code{eq:tet_gamma_lower_bound}}} \quad (130)$$

where $q(k)$ is any upper bound on the number of possible summands in the completeness relation Eq. (105) viewed as a function of k . As there less than or equal to $(s+1)^{r-1}$ Young diagrams with r rows with size at most s , letting $q(k)$ be given by Eq. (128) ensures Eq. (130) holds because the size (resp. rank) of any of $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ is at most $k\tau_{\max} + 3$ (resp. $r_{\max}$). Moreover, applying Eq. (111) yields

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k} \leq \frac{6}{k} \log(q(k)p(k)) \quad (131)$$

and thus Eq. (126) holds because $p(k)$ and $q(k)$ are polynomials in k with coefficients depending on the maximum trace $\tau_{\max}$ and maximum rank $r_{\max}$. Also Eq. (127) follows from Eq. (130) combined with the upper bound provided by Eq. (107). $\square$

Remark 42. The convergence condition above (Eq. (126)) in conjunction with the size conditions given by Eq. (110) together ensure that the Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converge to the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the following sense that

$$\begin{aligned} \frac{\alpha'_k}{k} &\rightarrow \alpha, & \frac{\beta'_k}{k} &\rightarrow \beta, & \frac{\gamma'_k}{k} &\rightarrow \gamma, \\ \frac{\delta'_k}{k} &\rightarrow \delta, & \frac{\epsilon'_k}{k} &\rightarrow \epsilon, & \frac{\phi'_k}{k} &\rightarrow \phi, \end{aligned} \quad \text{\textcolor{blue}{\{eq:sextuple_convergence\}}} \quad (132)$$

where $\frac{\lambda'_k}{k} \rightarrow \lambda$ is shorthand for $\lim_{k \rightarrow \infty} (\lambda'_k)_i / k = \lambda_i$ for all $i \in \{1, \dots, n\}$.

While the existence of a convergent sequence $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ of primed Young diagrams is necessary for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral by Corollary 41, our next result proves it is also sufficient. In other words, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral, then no such sequence can exist.

Corollary 43. *Suppose $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is a non-tetrahedral sextuple of positive semidefinite and non-zero eigenvalues with minimum trace $\tau_{\min}$. Then for all sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ indexed by $k > \tau_{\min}^{-1}$ with sizes given by Eq. (110) and converging to $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the sense that Eq. (126) holds, we have*

$$\left\| \left\{ \begin{smallmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \exp \left(-\frac{k\tau_{\min}}{12} D_*^{\alpha\beta\gamma\delta\epsilon\phi} + O(\log k) \right), \quad \text{\textcolor{blue}{\{eq:seq_ud_sizj_upper_bound\}}} \quad (133)$$

where $D_*^{\alpha\beta\gamma\delta\epsilon\phi} > 0$ is a positive constant which measures the distance from $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to the closed set of tetrahedral eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ with matching traces (i.e., $|\alpha| = |\tilde{\alpha}|$, $|\beta| = |\tilde{\beta}|$, etc.):

$$D_*^{\alpha\beta\gamma\delta\epsilon\phi} = \min_{\substack{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi} \\ \text{tetrahedral} \\ \forall \lambda: |\tilde{\lambda}| = |\lambda|}} \left(\|\underline{\alpha} - \tilde{\underline{\alpha}}\|_1^2 + \|\underline{\beta} - \tilde{\underline{\beta}}\|_1^2 + \|\underline{\gamma} - \tilde{\underline{\gamma}}\|_1^2 + \|\underline{\delta} - \tilde{\underline{\delta}}\|_1^2 + \|\underline{\epsilon} - \tilde{\underline{\epsilon}}\|_1^2 + \|\underline{\phi} - \tilde{\underline{\phi}}\|_1^2 \right). \quad (134)$$

Proof. Using Eq. (99), we have

$$R_{\alpha'\beta'\delta';\epsilon'} \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 = \iiint d\mu(A)d\mu(B)d\mu(D) \Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \quad (135)$$

where the integral is taken over a triple of $n \times n$ trace one positive semidefinite matrices A, B, D . For each integrand, define a tetrahedral sextuple $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$ as follows

$$\begin{aligned} \tilde{A} &:= |\alpha| A, & \tilde{B} &:= |\beta| B, & \tilde{C} &:= |\alpha| A + |\beta| B, \\ \tilde{D} &:= |\delta| D, & \tilde{E} &:= |\alpha| A + |\beta| B + |\delta| D, & \tilde{F} &:= |\beta| B + |\delta| D. \end{aligned} \quad \text{\textcolor{blue}{\{eq:tilde_defs\}}} \quad (136)$$

Furthermore, let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be the eigenvalues of $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$. In this notation, the integrand above is also equal to $\Gamma \left[\begin{smallmatrix} \tilde{A} & \tilde{B} & \tilde{C} \\ \tilde{D} & \tilde{E} & \tilde{F} \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right]$. Moreover, by Eq. (111) we have a bound on the integrand,

$$\left| \Gamma \left[\begin{smallmatrix} \tilde{A} & \tilde{B} & \tilde{C} \\ \tilde{D} & \tilde{E} & \tilde{F} \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \right| \leq p(k) \exp(-k I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} / 6). \quad (137)$$

Now by construction, the eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ are tetrahedral while by assumption, the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral (although they have matching traces due to Eq. (136)). Therefore we conclude that

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \neq (\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi}) \quad (138)$$

and therefore $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha\beta\gamma\delta\epsilon\phi} > 0$.

Our goal now will be to show that for any sequence of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converging to the non-tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$, one can also guarantee that for sufficiently large k , the exponent $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ above is also bounded from below by a positive constant. Let λ be a placeholder variable indexing $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$. By Pinsker's inequality for the relative entropy, we obtain a sequence of lower bounds on $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$:

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = \sum_{\lambda} |\lambda| D(\underline{\lambda}' \| \underline{\lambda}) \geq \frac{1}{2} \sum_{\lambda} |\lambda| \|\underline{\lambda}' - \underline{\lambda}\|_1^2 \geq \frac{\tau_{\min}}{2} \sum_{\lambda} \|\underline{\lambda}' - \underline{\lambda}\|_1^2 \geq \frac{\tau_{\min}}{2} \max_{\lambda} \|\underline{\lambda}' - \underline{\lambda}\|_1^2, \quad (139)$$

where $\tau_{\min}$ is the minimum trace over the eigenvalues (the lower indices). Applying the last bound to $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ yields

$$\max_{\lambda} \|\underline{\lambda}'_k - \underline{\lambda}\|_1 \leq \sqrt{2\tau_{\min}^{-1} I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}} \quad (140)$$

whereas applying the second to last bound to $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ (noting that $\tau_{\min}$ is also the minimum trace of $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$) yields

$$\sum_{\lambda} \|\underline{\lambda}'_k - \tilde{\lambda}\|_1^2 \leq 2\tau_{\min}^{-1} I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}. \quad (141)$$

Furthermore, by the reverse triangle inequality we have

$$\sum_{\lambda} \|\underline{\lambda}'_k - \tilde{\lambda}\|_1^2 \geq \sum_{\lambda} \left(\|\underline{\lambda}'_k - \underline{\lambda}\|_1 - \|\underline{\lambda} - \tilde{\lambda}\|_1 \right)^2 \geq \sum_{\lambda} \|\underline{\lambda} - \tilde{\lambda}\|_1^2 - 2 \|\underline{\lambda}'_k - \underline{\lambda}\|_1 \|\underline{\lambda} - \tilde{\lambda}\|_1. \quad (142)$$

Also by the definition of $D_{*}^{\alpha\beta\gamma\delta\epsilon\phi}$ we have

$$D_{*}^{\alpha\beta\gamma\delta\epsilon\phi} \leq \sum_{\lambda} \|\underline{\lambda} - \tilde{\lambda}\|_1^2. \quad (143)$$

Since $\underline{\lambda}$ and $\tilde{\lambda}$ are both probability distributions we also have $\|\underline{\lambda} - \tilde{\lambda}\|_1 \leq 2$. Also, $6 \max_{\lambda} \|\underline{\lambda}'_k - \underline{\lambda}\|_1 \geq \sum_{\lambda} \|\underline{\lambda}'_k - \underline{\lambda}\|_1$ as there are six summands. Combining all these bounds together yields

$$\left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \exp \left(-k \left\{ \frac{\tau_{\min} D_{*}^{\alpha\beta\gamma\delta\epsilon\phi}}{12} - 2 \sqrt{2\tau_{\min} I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}} - \frac{1}{k} \log(R_{\alpha'_k, \beta'_k, \delta'_k; \epsilon'_k}^{-1} p(k)) \right\} \right). \quad (144)$$

By assumption, the Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converge in normalization to the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the sense that $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ tends to zero as $k \rightarrow \infty$ at a rate given by Eq. (126). Moreover, $R_{\alpha'_k, \beta'_k, \delta'_k; \epsilon'_k}^{-1}$ can be upper bounded by a polynomial in k and $p(k)$ is a polynomial in k so Eq. (133) holds as claimed. $\square$

4 The large n limit & an entropic inequality

sec:large_n

The purpose of this section is to show how to use the necessary condition for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral provided by Corollary 41 to derive inequalities satisfied by tetrahedral eigenvalues. Our strategy will be to prove an upper bound the generalized $6j$ symbol which depends only on the dimensions of the underlying irreducible representations and the multiplicity spaces between them. By combining this upper bound with the lower bound provided by Corollary 41, we obtain a new constraint on the convergent sequence of Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$.

Theorem 44. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of Young diagrams. Then

$$\left\| \left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} \right\|_{HS}^2 \leq \left(c_{\alpha', \phi'}^{\epsilon'} c_{\beta', \delta'}^{\phi'} c_{\alpha', \beta'}^{\gamma'} c_{\gamma', \delta'}^{\epsilon'} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma'}^n) \dim(V_{\phi'}^n)}{\dim(V_{\beta'}^n) \dim(V_{\epsilon'}^n)}. \quad (145)$$

Proof. Here we provide a diagrammatic proof of the claimed inequality. For the sake of clarity, we abbreviate the irreducible representation labels, $V_{\lambda'}^n$, and omit the multiplicity space labels on the intermediate steps.

$$\left(\begin{array}{c} \text{Diagrammatic expression} \end{array} \right)^{\frac{1}{2}} = \frac{\left(\begin{array}{c} \text{Diagrammatic expression} \end{array} \right)^{\frac{1}{2}}}{\left(\begin{array}{c} \text{Diagrammatic expression} \end{array} \right)^{\frac{1}{2}}} \quad (146)$$

The first equality follows from Definition 36. The inequality follows from an application of the Cauchy-Schwarz inequality with respect to the annotated contraction. The final equality follows from applying Schur's lemma to components of the diagram mapping $V_{\beta'}^n$ to itself on the left and $V_{\epsilon'}^n$ to itself on the right. Note that the factors of $\dim(V_{\epsilon'}^n)^2$ appearing on left-most and right-most sides of the above inequality cancel producing the claimed inequality. $\square$

An immediate application of Theorem 44 is to combine it with Corollary 41. If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral eigenvalues, then by Corollary 41 there exists a converging sequence of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ such that⁸

$$\frac{1}{q(k)^2} \leq \left\| \left\{ \begin{array}{ccc} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{array} \right\} \right\|_{HS}^2 \leq \left(c_{\alpha'_k, \phi'_k}^{\epsilon'_k} c_{\beta'_k, \delta'_k}^{\phi'_k} c_{\alpha'_k, \beta'_k}^{\gamma'_k} c_{\gamma'_k, \delta'_k}^{\epsilon'_k} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma'_k}^n) \dim(V_{\phi'_k}^n)}{\dim(V_{\beta'_k}^n) \dim(V_{\epsilon'_k}^n)}. \quad (147)$$

Remarkably, both $q(k)$ (given by Eq. (128)) and the Littlewood-Richardson coefficients $c_{\alpha'_k, \phi'_k}^{\epsilon'_k}$, $c_{\beta'_k, \delta'_k}^{\phi'_k}$, $c_{\alpha'_k, \beta'_k}^{\gamma'_k}$, and $c_{\gamma'_k, \delta'_k}^{\epsilon'_k}$ are independent of the dimension parameter n defining the group $U(n)$ (assuming n is at least as large as the maximum depth of any of the involved Young diagrams, i.e., $n \geq r_{\max}$ where $r_{\max}$ is the maximum rank of any of the tetrahedral eigenvalues). This observation provides one with an opportunity to optimize the above bound by varying the value of n .

In particular, it will be useful to better understand how the dimension, $\dim(V_{\lambda'}^n)$, of the irreducible representation $V_{\lambda'}^n$ of $U(n)$ grows as n becomes much larger than the number of rows of λ' . In general, for fixed λ' , the function $n \mapsto \dim(V_{\lambda'}^n)$ can grow at an exponential rate with increasing n . One can find estimates for $\dim(V_{\lambda'}^n)$ in the regime where n is large compared to $|\lambda'|$ and also the last row $\lambda'_{\min}$ is large compared to the depth of λ' in paper by Itzykson and Nauenberg [IN66, Section III]. Here we use an elegant presentation of this scaling, namely [Har23, Lemma 2], which relates the asymptotics of $\dim(V_{\lambda'}^n)$ to $\dim(W_{\lambda'})$.

Proposition 45. Let λ' be a Young diagram with depth at most n . Then

$$\frac{\dim(V_{\lambda'}^n)}{n^{|\lambda'|}} = \frac{\dim(W_{\lambda'})}{|\lambda'|!} \prod_{(i,j) \in \lambda'} \left(1 + \frac{j-i}{n} \right). \quad (148)$$

Proof. The statement follows from the dimension formulas for $\dim(V_{\lambda'}^n)$ and $\dim(W_{\lambda'})$ in Eq. (35) and Eq. (36). See also [Har23, Appendix A]. $\square$

From Proposition 45 we obtain the following corollary.

⁸Here we used the well-known inequality $\|X\|_{\infty} \leq \|X\|_{HS}$.

Corollary 46. *Let $\gamma', \phi', \beta', \epsilon'$ be a quadruple of Young diagrams such that*

$$|\gamma'| + |\phi'| = |\beta'| + |\epsilon'|. \quad (149)$$

Then

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\gamma'}^n) \dim(V_{\phi'}^n)}{\dim(V_{\beta'}^n) \dim(V_{\epsilon'}^n)} = \frac{|\beta'|! |\epsilon'|! \dim(W_{\gamma'}) \dim(W_{\phi'})}{|\gamma'|! |\phi'|! \dim(W_{\beta'}) \dim(W_{\epsilon'})} \quad (150)$$

Proof. From Proposition 45 one evidently has for any Young diagram λ' the limit

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\lambda'}^n)}{n^{|\lambda'|}} = \frac{\dim(W_{\lambda'})}{|\lambda'|!}. \quad (151)$$

The claim then follows because the missing prefactor, $\frac{n^{|\beta'|} n^{|\epsilon'|}}{n^{|\gamma'|} n^{|\phi'|}}$, is equal to one by Eq. (149). $\square$

We are now in a position to prove the main result of this section.

Theorem 47. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be tetrahedral eigenvalues. Then*

$$I(\gamma, \phi; \beta, \epsilon) = s(\gamma) + s(\phi) - s(\beta) - s(\epsilon) \geq 0, \quad (152)$$

where

$$s(\lambda) := - \sum_{i=1}^n \lambda_i \log \lambda_i. \quad (153)$$

Proof. As $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral, the trace conditions (Eq. (13)) necessarily imply that Eq. (149) holds. Therefore, by Corollary 46 and the discussion surrounding the bound in Eq. (147) we conclude that along the convergent sequence of Young diagrams satisfies

$$I' := \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[\frac{|\beta'_k|! |\epsilon'_k|! \dim(W_{\gamma'_k}) \dim(W_{\phi'_k})}{|\gamma'_k|! |\phi'_k|! \dim(W_{\beta'_k}) \dim(W_{\epsilon'_k})} \right] \geq 0 \quad (154)$$

provided the limit I' exists. An application of Stirling's approximation [Rob55] to the dimension formula for $\dim(W_{\lambda'_k})$ together with the approximation $\lambda'_k \approx k\lambda$ yields

$$\frac{1}{k} \log \left(\frac{\dim(W_{\lambda'_k})}{|\lambda'|!} \right) = s(\lambda) + |\lambda| (1 - \log k) + O(k^{-1} \log k) \quad (155)$$

where $s(\lambda)$ is given in Eq. (153). Moreover by Eq. (149), the $(1 - \log k)$ terms cancel and therefore $I' = I(\gamma, \phi; \beta, \epsilon) \geq 0$ as claimed above. $\square$

Acknowledgements

MC and TC acknowledge financial support from the European Research Council (ERC Grant Agreement No. 818761), VILLUM FONDEN via the QMATH Centre of Excellence (Grant No. 10059) and the Novo Nordisk Foundation (grant NNF20OC0059939 ‘Quantum for Life’). They also thank the National Center for Competence in Research SwissMAP of the Swiss National Science Foundation and the Section of Mathematics at the University of Geneva for their hospitality.

A On the oriented tetrahedral symmetry

The group G of order 48 appearing in Proposition 8 corresponds to the subgroup of all possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple

(A, B, C, D, E, F) which preserves the set of linear constraints in Eq. (7). The group G is isomorphic to both the direct product $S_2 \times S_4$ and to the signed permutation group of three elements, i.e., $(S_2 \times S_2 \times S_2) \rtimes S_3$.

The first isomorphism, $G \simeq S_2 \times S_4$, can be understood as S_2 acting by the inversion of all eigenvalues,

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \leftrightarrow (\alpha^*, \beta^*, \gamma^*, \delta^*, \epsilon^*, \phi^*), \quad (156)$$

while S_4 acts by permutation of the *vertices* of the tetrahedron in Fig. 1: each vertex permutation induces a permutation of the six directed edges with the possibility of inverting the orientation of a subset of the edges (such edge orientation reversals can be accounted for a corresponding inversion of eigenvalues).

The second isomorphism, $G \simeq (S_2 \times S_2 \times S_2) \rtimes S_3$, can be understood by considering the triplet $(A, D), (B, E), (C, F)$ of non-adjacent pairs of edges in Fig. 1. Accordingly, it is helpful to arrange the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ into a 2×3 grid using the following identification:

$$(\alpha \ \beta \ \gamma \ \delta \ \epsilon \ \phi) \cong \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix}. \quad (157)$$

Then one can see that each S_2 factor in the direct product $S_2 \times S_2 \times S_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \alpha & \epsilon^* & \phi^* \\ \delta & \beta^* & \gamma^* \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta & \beta & \phi \\ \alpha & \epsilon & \gamma \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta^* & \epsilon & \gamma \\ \alpha^* & \beta & \phi \end{pmatrix}. \quad (158)$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \quad \begin{pmatrix} \beta & \gamma^* & \alpha^* \\ \epsilon & \phi & \delta \end{pmatrix} \quad \begin{pmatrix} \gamma^* & \alpha & \beta^* \\ \phi & \delta & \epsilon \end{pmatrix} \\ \begin{pmatrix} \alpha^* & \gamma & \beta \\ \delta & \phi & \epsilon \end{pmatrix} \quad \begin{pmatrix} \beta^* & \alpha^* & \gamma^* \\ \epsilon & \delta & \phi \end{pmatrix} \quad \begin{pmatrix} \gamma & \beta^* & \alpha \\ \phi & \epsilon & \delta \end{pmatrix} \quad (159)$$

We further remark that these symmetries also corresponding to symmetries of the Wigner $6j$ operator $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ in Definition 36 with the involution $\lambda \mapsto \lambda^*$ corresponding taking the dual representation $V_\lambda^n \mapsto (V_\lambda^n)^*$. In the case where $n = 2$, the involution map becomes the identity map (as every irreducible representation of $SU(2)$ is self-dual, or the reversal of a pair $(x, -x)$ of traceless eigenvalues is trivial), and thus the symmetry group collapses to a group of order $|S_4| = 4! = 24$.

B Multinomial distribution inequalities

lem:multinomial_dist_lower_bound

Lemma 48. Fix $m \geq 2$ and let $(w_1, \dots, w_m) \in \mathbb{R}_{>0}^m$ have minimum $w_{\min} := \min\{w_1, \dots, w_m\} > 0$. Let k be a positive integer satisfying $k \geq w_{\min}^{-1}$ and define positive integers $(z_1, \dots, z_m) \in \mathbb{N}_{>0}^m$ by the equation

$$z_i := \lceil kw_i \rceil. \quad (160)$$

Let $Z = z_1 + \dots + z_m$ and $W = w_1 + \dots + w_m$. Then

$$\frac{Z!}{z_1! \dots z_m!} \frac{w_1^{z_1} \dots w_m^{z_m}}{W^Z} \geq \left(\frac{1}{12} \right)^m (kW + m)^{\frac{m-1}{2}} \quad \text{\textit{eq:multinomial_dist_lower_bound}} \quad (161)$$

Proof. The proof follows from a version of Stirling's approximation due to Robbins [Rob55] which states

$$k! = \sqrt{2\pi k} k^{k+\frac{1}{2}} e^{-k} e^{r_k}, \quad \text{where} \quad \frac{1}{12k+1} \leq r_k \leq \frac{1}{12k}. \quad (162)$$

Therefore,

$$\frac{Z!}{z_1! \dots z_m!} \frac{w_1^{z_1} \dots w_m^{z_m}}{W^Z} = \left(\frac{1}{2\pi Z} \right)^{\frac{m-1}{2}} \prod_{i=1}^m \left(\frac{Z}{z_i} \right)^{\frac{1}{2}} \left(\frac{Z w_i}{z_i W} \right)^{z_i} \exp \left(r_Z - \sum_{i=1}^m r_{z_i} \right). \quad (163)$$

In this setting, we have let $z_i = \lceil kw_i \rceil$ which implies

$$kw_i \leq z_i < kw_i + 1, \quad \text{and} \quad kW \leq Z < kW + m. \quad (164)$$

Consequently⁹,

$$\left(\frac{Zw_i}{z_i W} \right)^{z_i} \geq \left(\frac{kw_i}{kw_i + 1} \right)^{kw_i + 1} \geq \frac{1}{4}. \quad (165)$$

The remainder term satisfies

$$r_Z - \sum_i r_{z_i} \geq \frac{1}{12Z + 1} - \sum_i \frac{1}{12z_i} \geq -\frac{m}{12}. \quad (166)$$

Also $Z \geq z_i$ so altogether, we have

$$\frac{Z!}{z_1! \cdots z_m!} \frac{w_1^{z_1} \cdots w_m^{z_m}}{W^Z} \geq \left(\frac{1}{2\pi} \right)^{\frac{m-1}{2}} \left(\frac{1}{4} \right)^m \exp\left(-\frac{m}{12}\right) \left(\frac{1}{kW + m} \right)^{\frac{m-1}{2}} \geq \left(\frac{1}{12} \right)^m (kW + m)^{\frac{m-1}{2}}, \quad (167)$$

as claimed because $\sqrt{2\pi} > 1$ and $\frac{1}{12} < \frac{e^{-\frac{1}{12}}}{4\sqrt{2\pi}} = 0.09176 \dots$. $\square$

C A character formula for $6j$ symbols

Throughout Section 3.3 we proved a number of results concerning the square of the Hilbert-Schmidt norm of the $6j$ operator defined by Definition 36. The following result provides a formula for the square of the Hilbert-Schmidt norm as an integral over four copies of $U(n)$.

lem:character_formula

Lemma 49 (Character formula). *Let $\left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} : C_{\alpha', \beta'}^{\gamma'} \otimes C_{\delta', \gamma'}^{\epsilon'} \rightarrow C_{\alpha', \phi'}^{\epsilon'} \otimes C_{\beta', \delta'}^{\phi'}$ be the $6j$ operator defined in Definition 36. Then*

$$\dim(V_{\epsilon'})^2 \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 = \int_{U(n)^4} du_1 du_2 du_3 du_4 X_{\delta', \epsilon', \phi'}^{\alpha', \beta', \gamma'}(u_1, u_2, u_3, u_4). \quad (168)$$

eq:character_formula

where

$$X_{\delta', \epsilon', \phi'}^{\alpha', \beta', \gamma'}(u_1, u_2, u_3, u_4) := \chi_{\alpha'}(u_1^{-1}u_2) \chi_{\beta'}(u_4^{-1}u_2) \chi_{\gamma'}(u_2^{-1}u_3) \chi_{\delta'}(u_4^{-1}u_3) \chi_{\epsilon'}(u_3^{-1}u_1) \chi_{\phi'}(u_1^{-1}u_4). \quad (169)$$

Note that when $n = 2$, Eq. (168) recovers the well-known formula for the square of the $6j$ -symbol for $SU(2)$ (see [FL03, Equation (1)] and also [Bar98; FK00]).

Proof of Lemma 49. The proof follows from (i) the following identities concerning the character $\chi_{\lambda'}(u) = \text{Tr}(\pi_{\lambda'}(u))$ and duality $\pi_{\lambda'}^*(u) = (\pi_{\lambda'}(u^{-1}))^T$ relations

$$\chi_{\lambda}(u) = \boxed{\pi_{\lambda}(u)} \quad \pi_{\lambda}^*(u) = \boxed{\pi_{\lambda}(u^{-1})}, \quad (170)$$

and (ii) the integral formula for the projection operator $\Pi_{\alpha, \beta}^{\gamma}$ onto the subspace of invariant vectors in $V_{\alpha} \otimes V_{\beta} \otimes V_{\gamma}^*$:

$$\Pi_{\alpha, \beta}^{\gamma} = \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \downarrow \\ \boxed{\Pi_{\alpha, \beta}^{\gamma}} \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array} = \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \downarrow \\ \circ \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array} C_{\alpha, \beta}^{\gamma} = \int du \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \downarrow \\ \boxed{\pi_{\alpha}(u)} \quad \boxed{\pi_{\beta}(u)} \quad \boxed{\pi_{\gamma}^*(u)} \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array}. \quad (171)$$

⁹Here we use the identity that $x \geq 1$ implies $\left(\frac{x}{x+1} \right)^{x+1} \geq \frac{1}{4}$.

Applied four times, we obtain Eq. (168):

(172)

□

We conclude by noting that the integrands appearing in the above character formula are closely related to the character formula for the heat kernel of a compact Lie group G [Feg78], denoted by $K_G(u_1, u_2; t)$ in [McS21]:

$$K_G(u_1, u_2; t) = \sum_{\lambda'} \dim V_{\lambda'}^n \chi_{\lambda'}(u_1 u_2^{-1}) e^{-t(|\lambda' + \rho|^2 - |\rho|^2)/2}, \quad (173)$$

where the sum is over all dominant integral weights λ' for G and ρ the associated Weyl vector. It is conceivable that the square of the Hilbert-Schmidt norm of the generalized $6j$ symbols for G can be seen as a low temperature limit, $t \rightarrow 0$, of an integral over products of heat kernels.

D Seesaw reciprocity & eigenvalues of blocks

sec:blocks_of_psd_matrices

The purpose of this section is to reformulate both Horn's problem and tetrahedral Horn problem as a instance of a slightly more general kind of eigenvalue problem involving positive semidefinite matrices. Throughout this section, we will restrict our focus to the case of positive semidefinite matrices and their eigenvalues. As a reminder we note that this can be done without loss of generality due to Corollary 11.

Given a positive semidefinite matrix, X , with eigenvalues $\text{eig}(X) = \chi = (\chi_1 \geq \dots \geq \chi_n \geq 0)$ and eigen-decomposition $X = \sum_{i=1}^n \chi_i \Pi_i$, one can define $X^{\frac{1}{2}}$ by $X^{\frac{1}{2}} := \sum_{i=1}^n \chi_i^{\frac{1}{2}} \Pi_i$. Inspired by [Bha97, Theorem IV.2.13], we obtain the following fundamental lemma relating the eigenvalues γ of a sum $C = A + B$ of $n \times n$ Hermitian matrices to the eigenvalues of a larger $2n \times 2n$ matrix M built from A and B .

lem:trigonal_block

Lemma 50. *Let (A, B, C) be a triple of $n \times n$ Hermitian matrices with eigenvalues (α, β, γ) such that $C = A + B$. Let C' be the $2n \times 2n$ Hermitian matrix*

$$C' = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B \end{bmatrix}. \quad (174)$$

Then $C' \geq 0$ is positive semidefinite, has rank at most n and has eigenvalues given by γ

$$\text{eig}(C') \simeq \text{eig}(A + B) = \text{eig}(C) = \gamma. \quad (175)$$

Proof. That C' is positive semidefinite and has rank at most n follows because C' can be decomposed as $C' = T^* T$ where T is the $n \times 2n$ block matrix

$$T = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix}. \quad (176)$$

That C' and $A+B$ share the same eigenvalues (up to padding by n zeros) follows because for any rectangular matrix T , the eigenvalues of T^*T and TT^* are equal (up to padding by zeros) and in this instance

$$T^*T = \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}}B^{\frac{1}{2}} \\ B^{\frac{1}{2}}A^{\frac{1}{2}} & B \end{bmatrix} = C' \quad TT^* = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} = A+B. \quad (177)$$

□

From the perspective of Lemma 50, one can view the Horn's problem involving the eigenvalues (α, β, γ) of $n \times n$ Hermitian matrices (A, B, C) satisfying $A+B=C$ as a special case of a more general eigenvalue problem concerned with the relations between the eigenvalues of $2n \times 2n$ Hermitian matrices Σ of the form

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad \text{\texttt{eq:block_ABX}} \quad (178)$$

and the eigenvalues of its two $n \times n$ diagonal blocks Z and Z . It is known, by [HJ12, Theorem 7.7.9] or more generally [MKX19, Theorem 5.11] that if C' is a 2×2 block matrix such that $A', B' \geq 0$, then $C' \geq 0$ if and only if the off-diagonal block Z is of the form

$$Z = A'^{\frac{1}{2}}KB'^{\frac{1}{2}} \quad (179)$$

for some contraction matrix K (largest eigenvalue of $(K^*K)^{\frac{1}{2}}$ is at most unity). Additionally, when $Z = A'^{\frac{1}{2}}KB'^{\frac{1}{2}}$, one has the decomposition

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix} = \begin{bmatrix} A'^{\frac{1}{2}} & 0 \\ 0 & B'^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} I & K \\ K^* & I \end{bmatrix} \begin{bmatrix} A'^{\frac{1}{2}} & 0 \\ 0 & B'^{\frac{1}{2}} \end{bmatrix}, \quad (180)$$

noting that K is a contraction if and only if the middle block matrix above is positive semidefinite [MKX19, Lemma 5.10].

Therefore, the special case covered by Lemma 50 corresponds to the case where the corresponding contraction matrix K is the $n \times n$ identity matrix $K = I$ (or more generally any contractive matrix satisfying $A'^{\frac{1}{2}}KB'^{\frac{1}{2}} = A'^{\frac{1}{2}}B'^{\frac{1}{2}}$). A simple observation which distinguishes between eigenvalue problem concerning diagonal blocks and the eigenvalue problem concerning sums is simply that the rank of C' can in general be up to $2n$ whereas the rank of the sum $C = A+B$ is at most n . Despite this additional freedom, there remains non-trivial relations constraining the eigenvalues of C' and of the blocks A' and B' in terms of properties of Z (see [Gum+18; BL22]). Nevertheless, when the rank of C' is constrained to be at most n , we obtain the following theorem which provides an equivalent formulation of Horn's problem.

\texttt{thm:trigonal_block}

Theorem 51. *Let (α, β, γ) be a triple of positive semidefinite eigenvalues of length at most n . The following are equivalent:*

- (i) *there exists a triple of Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A+B=C$,*
- (ii) *there exists an $2n \times 2n$ Hermitian matrix,*

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad (181)$$

such that $\text{eig}(A') = \alpha$, $\text{eig}(B') = \beta$, and $\text{eig}(C') = \gamma$.

Proof. Since we explicitly assume all of the eigenvalues in the triple (α, β, γ) are non-negative, the Hermitian matrices A, B, C , and A', B', C' involved in the statements above are necessarily positive semidefinite. The claim (i) $\implies$ (ii) follows from Lemma 50. That (ii) $\implies$ (i) follows by noticing that if C' has eigenvalues γ , then it has rank at most n . Therefore, as $C' \geq 0$, there exists a $n \times 2n$ block matrix T of the form

$$T = \begin{bmatrix} T_A & T_B \end{bmatrix} \quad (182)$$

such that $C' = T^*T$. The diagonal blocks of C' are therefore $A' = T_A^*T_A$, and $B' = T_B^*T_B$. As in the proof of Lemma 50, we note that T^*T and TT^* have the same eigenvalues up to padding by zeros. Therefore, by letting $A = T_A T_A^*$ and $B = T_B T_B^*$ we see that

$$\begin{aligned} \text{eig}(A) &= \text{eig}(T_A T_A^*) \simeq \text{eig}(T_A^* T_A) = \text{eig}(A') = \alpha, \\ \text{eig}(B) &= \text{eig}(T_B T_B^*) \simeq \text{eig}(T_B^* T_B) = \text{eig}(B') = \beta, \end{aligned} \quad (183)$$

while the eigenvalues of $C = A + B = T_A T_A^* + T_B T_B^*$ match the eigenvalues of C' because

$$\text{eig}(C) = \text{eig}(A + B) = \text{eig}(T_A T_A^* + T_B T_B^*) = \text{eig}(TT^*) \simeq \text{eig}(T^*T) = \text{eig}(C') = \gamma. \quad (184)$$

□

Here we established a correspondence between (i) the existence of $n \times n$ positive semidefinite matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$ and (ii) the positivity of corresponding Littlewood-Richardson coefficients $c_{\alpha\beta}^\gamma$ which govern the branching problem for the diagonal subgroup $\text{GL}(n) \hookrightarrow \text{GL}(n) \times \text{GL}(n)$ sending X to (X, X) . Recalling Theorem 51, there is a further correspondence to (iii) the existence of a $2n \times 2n$ positive semidefinite matrix Σ with eigenvalues γ and $n \times n$ blocks A', B' with eigenvalues α, β . One may wonder how to show the correspondence between (ii) and (iii) *directly* without passing through (i). This can be accomplished by considering the branching multiplicities $m_\gamma^{\alpha\beta}$ for the blockwise subgroup $\text{GL}(a) \times \text{GL}(b) \hookrightarrow \text{GL}(a+b)$ sending (A, B) to $A \oplus B$. In fact there exists a representation-theoretic analogue of Theorem 51, specifically [GW00, Theorem 9.2.3]. In particular, the values of $m_\gamma^{\alpha\beta}$ and $c_{\alpha\beta}^\gamma$ coincide provided the Young diagrams (α, β, γ) involved have depths at most $(\min\{a, n\}, \min\{b, n\}, \min\{a+b, n\})$; this ensures that $V_\alpha^a, V_\beta^b, V_\gamma^{a+b}$ and $V_\alpha^n, V_\beta^n, V_\gamma^n$ are all non-zero representation spaces and therefore $m_\gamma^{\alpha\beta}$ and $c_{\alpha\beta}^\gamma$ are both meaningfully defined. The distinction between $m_\gamma^{\alpha\beta}$ and $c_{\alpha\beta}^\gamma$ lies with the fact that it is possible for $c_{\alpha\beta}^\gamma$ to be zero while $m_\gamma^{\alpha\beta}$ is non-zero (this can happen when the depth of γ is greater than n but at most $a+b$ so that V_γ^n is zero but V_γ^{a+b} is non-zero). This parallels the fact that, in general, the rank of the $2n \times 2n$ matrix Σ in Theorem 51 could be greater than n while the rank of the $n \times n$ matrix C is at most n .

As a corollary of this correspondence between the Horn's problem and the eigenvalues of blocks of 2×2 matrices, we obtain an equivalent formulation of the tetrahedral Horn problem. thm:tetrahedral_block

Theorem 52. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of positive semidefinite eigenvalues of length at most n . Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if there exists a $3n \times 3n$ matrix E' of the form*

$$E' = \begin{bmatrix} A' & Z & Y \\ Z^* & B' & X \\ Y^* & X^* & D' \end{bmatrix}, \quad \text{with blocks} \quad C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad F' = \begin{bmatrix} B' & X \\ X^* & D' \end{bmatrix} \quad \text{eq:block_3by3} \quad (185)$$

such that

$$\text{eig}(A') = \alpha, \quad \text{eig}(B') = \beta, \quad \text{eig}(C') = \gamma, \quad \text{eig}(D') = \delta, \quad \text{eig}(E') = \epsilon, \quad \text{eig}(F') = \phi. \quad \text{eq:block_eigenvalue_tet} \quad (186)$$

Proof. The proof is analogous to the proofs of Lemma 50 and Theorem 51. First we prove the “only if” portion of the claim which is a corollary of Lemma 50. Suppose (A, B, C, D, E, F) is a tetrahedral sextuple of Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \gamma)$. Let E' be given by

$$E' = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix}^* \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} & A^{\frac{1}{2}} D^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B & B^{\frac{1}{2}} D^{\frac{1}{2}} \\ D^{\frac{1}{2}} A^{\frac{1}{2}} & D^{\frac{1}{2}} B^{\frac{1}{2}} & D \end{bmatrix} \quad (187)$$

Then the eigenvalues of the corresponding principal submatrices of E' satisfy Eq. (186) by Lemma 50. Now we prove the “if” portion of the claim which is a corollary of Theorem 51. If E' is a matrix of form given by Eq. (185) with eigenvalues ϵ with length at most n , then E' is positive semidefinite with rank at most n . Therefore, there exists an $n \times 3n$ block matrix

$$T = \begin{bmatrix} T_A & T_B & T_D \end{bmatrix} \quad (188)$$

such that $E' = T^*T$ and thus

$$\begin{aligned} A' &= T_A^* T_A, & B' &= T_B^* T_B, & D' &= T_D^* T_D, \\ X &= T_B^* T_D, & Y &= T_A^* T_D, & Z &= T_A^* T_B. \end{aligned} \quad (189)$$

Now let (A, B, C, D, E, F) be given by

$$\begin{aligned} A &= T_A T_A^*, & B &= T_B T_B^*, & C &= T_A T_A^* + T_B T_B^*, \\ D &= T_D T_D^*, & E &= T_A T_A^* + T_B T_B^* + T_D T_D^*, & F &= T_B T_B^* + T_D T_D^*. \end{aligned} \quad (190)$$

From here, one may directly verify that these $n \times n$ Hermitian matrices satisfy Eq. (7) and have eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ (using a similar argument presented in the proof of Theorem 51); i.e., one concludes that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral. $\square$

TODO: Describe the same correspondence for the blockwise $6j$ symbols.

E Inequalities for tetrahedral eigenvalues

sec:ineqs

The purpose of this section is to describe various inequalities which are necessarily satisfied by tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. In general, however, the task of finding a complete description of the space of tetrahedral eigenvalues for $n \geq 3$ remains an open problem.

E.1 Horn's inequalities

Recall that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral only if each of (α, β, γ) , (β, γ, ϕ) , $(\gamma, \delta, \epsilon)$, and (α, ϕ, ϵ) are trigonal as in Eq. (9). Consequently, every inequality constraining trigonal eigenvalues can be lifted to produce a quadruplet of inequalities constraining the set of tetrahedral eigenvalues. These inequalities can be understood as generalizations of the triangle inequalities, Eq. (17), which hold for case where $n = 2$. Therefore, before proceeding, we briefly summarize the inequalities satisfied by trigonal eigenvalues (α, β, γ) . Of particular interest are Horn's inequalities [Hor62] which are known to be necessary and sufficient [Kly98; KT99]. For a more complete historical account, the reader is directed to the survey articles [Bha01; Ful98; Knu00] or [NP06, Appendix D]. In addition to the aforementioned trace condition, $\sum_{i=1}^n \alpha_i + \sum_{j=1}^n \beta_j = \sum_{k=1}^n \gamma_k$, one always has the following pair of elementary inequalities,

$$\alpha_1 + \beta_1 \geq \gamma_1, \quad \text{and} \quad \alpha_n + \beta_n \leq \gamma_n, \quad (191)$$

which constrain the largest and smallest eigenvalues respectively (note that under inversion symmetry, these inequalities are equivalent). The first somewhat non-trivial inequality, derived by Weyl in 1912, is now known as *Weyl's inequality* and states for all i, j such that $i + j \leq n + 1$ [Wey12]¹⁰

$$\alpha_i + \beta_j \geq \gamma_{i+j-1}. \quad (192)$$

In 1949, Fan derived a majorization inequality constraining the first r largest eigenvalues [Fan49], and shortly thereafter Lidskii [Lid53] & Wielandt [Wie55] found proofs of another family of inequalities. Finally, in 1962, Horn determined a complete collection of inequalities for (α, β, γ) are trigonal for the case where $n = 3$ [Hor62]. Moreover, also in [Hor62], Horn provided an inductively defined set of subsets $I, J, K \subset \{1, \dots, n\}$ of equal cardinality and conjectured that inequalities of the form

$$\sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j \geq \sum_{k \in K} \gamma_k, \quad \text{\texttt{\{eq:IJK_ineq\}}} \quad (193)$$

were both necessary and sufficient to ensure that (α, β, γ) are trigonal. While proofs for some of these inequalities were found (see e.g., [TF71; TF70]), it took until the late 1990s for Horn's problem to be to

¹⁰For 2×2 matrices one can see that these necessary inequalities are also sufficient to ensure that (α, β, γ) are trigonal; upon shifting to traceless eigenvalues using Corollary 10, $\alpha = (a, -a)$, $\beta = (b, -b)$ and $\gamma = (c, -c)$, and these inequalities become equivalent to the triangle inequalities $c \leq a + b$ and $c \geq \max\{b - a, a - b\}$.

be solved by Klyachko [Kly98] who proved that (α, β, γ) if and only if in addition to the trace equality, $|\alpha| + |\beta| = |\gamma|$, Eq. (193) holds for all (I, J, K) such that (i) $|I| = |J| = |K| = p$ for some p , and (ii) the Littlewood-Richardson coefficient $c_{\alpha', \beta'}^{\gamma'}$ is non-zero (here $\alpha' = (i_p - p, \dots, i_1 - 1)$, $\beta' = (j_p - p, \dots, j_1 - 1)$, and $\gamma' = (k_p - p, \dots, k_1 - 1)$). Shortly thereafter, Knutson and Tao proved the saturation conjecture for Littlewood-Richardson coefficients [KT99], which when combined with Klyachko's solution, proved Horn's conjecture.

For the purposes of characterizing the set of tetrahedral eigenvalues further, we will henceforth be exclusively concerned with inequalities constraining the set of tetrahedral eigenvalues which are not already implied by Horn's inequalities.

E.2 A simple quadratic inequality

In Section 2.3 we saw that the shifting symmetry implies, in a certain sense, one may assume without loss of generality that the candidate eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are all non-negative or even strictly positive. While almost all of the following inequalities require this positivity assumption, we note that they can be extended to eigenvalues with negative entries in all cases.

To begin, we provide an example of a non-linear inequality constraining tetrahedral eigenvalues which appears to be new.

prop:simple_quadratic

Proposition 53. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be non-negative tetrahedral eigenvalues of $n \times n$ Hermitian matrices. Then*

$$2 \sum_{j=1}^n \alpha_j \delta_{n-j+1} \leq \sum_{j=1}^n (\beta_j^2 + \epsilon_j^2 - \gamma_j^2 - \phi_j^2) \leq 2 \sum_{j=1}^n \alpha_j \delta_j. \quad (194)$$

Proof. Let (A, B, C, D, E, F) be the tetrahedral sextuple of positive semidefinite matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ satisfying $A + B = C$, $B + D = F$, and $A + F = C + D = E$. Then the following identity holds:

$$\begin{aligned} & \text{Tr}[B^2] + \text{Tr}[E^2] - \text{Tr}[C^2] - \text{Tr}[F^2] \\ &= \text{Tr}[(B)^2] + \text{Tr}[(A + B + D)^2] - \text{Tr}[(A + B)^2] - \text{Tr}[(B + D)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[AB] + 2 \text{Tr}[BD] + 2 \text{Tr}[AD]) \\ &\quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2 \text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[BD]) \\ &= 2 \text{Tr}[AD]. \end{aligned} \quad (195)$$

{eq:quad_identity}

At this stage, one may note that for all $A, D \geq 0$ positive semidefinite, $\text{Tr}(AD) \geq 0$ and therefore,

$$\text{Tr}[B^2] + \text{Tr}[E^2] - \text{Tr}[C^2] - \text{Tr}[F^2] \geq 0. \quad (196)$$

{eq:quadratic_difference_pos}

This inequality is well-known and indeed follows as a special of [Sag13, Proposition 2.3]. The improvement made here is to note that while the value of $\text{Tr}(AD)$ cannot, in general, be determined from the eigenvalues $\alpha = \text{eig}(A)$ and $\delta = \text{eig}(D)$, one can apply the von Neumann trace inequalities ([MOA10, H.1.g & H.1.h] or [Bau+01, Proposition 4.4]):

$$\sum_j \alpha_j \delta_{n-j+1} \leq \text{Tr}[AD] \leq \sum_j \alpha_j \delta_j, \quad (197)$$

where $\delta_{n-j+1} = -\delta_j^*$. The claim then follows because $\text{Tr}[X^2] = \sum_j \lambda_j^2$ for any Hermitian operator X with eigenvalues $\text{eig}(X) = \lambda$. $\square$

E.3 Ptolemy & Cayley-Menger inequalities

In this section, we prove inequalities constraining the 2-norms $\|\cdot\|_2$ of a tetrahedral sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ associated to the Hermitian matrices (A, B, C, D, E, F) . These inequalities correspond to Ptolemy's inequality and the Cayley-Menger determinant inequality for the edge-lengths of a Euclidean tetrahedron in $\mathbb{R}^3$. The core idea is to consider the Hilbert-Schmidt inner product $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$ on $n \times n$ Hermitian matrices and the induced Hilbert-Schmidt norm $\|X\| = \sqrt{\langle X, X \rangle}$. For any Hermitian matrix

X with eigenvalues λ , we have $\|X\| = \|\lambda\|_2 = \left(\sum_{j=1}^n \lambda_j^2\right)^{\frac{1}{2}}$. Recall that the four vertices of the tetrahedron of Hermitian matrices are

$$V_0 = 0, \quad V_1 = A, \quad V_2 = A + B = C, \quad V_3 = A + B + D = E. \quad (198)$$

E.3.1 Ptolemy's Inequality

Ptolemy's inequality applied any four points V_0, V_1, V_2, V_3 in a normed space reads

$$\|V_3 - V_0\| \|V_1 - V_2\| \leq \|V_2 - V_0\| \|V_1 - V_3\| + \|V_1 - V_0\| \|V_2 - V_3\|. \quad (\text{eq:vvvv-ptolemy}) \quad (199)$$

Applied to the vertices of the tetrahedron of Hermitian matrices, we obtain

$$\|E\| \|B\| \leq \|C\| \|F\| + \|A\| \|D\|. \quad (\text{eq:abcdef-ptolemy}) \quad (200)$$

It is known that every real inner product space with norm $\|X\| = \sqrt{\langle X, X \rangle}$ is Ptolemaic in the sense that Ptolemy's inequality holds. A converse of this result also holds due to Schoenberg; namely, every real semi-normed space satisfying Ptolemy's inequality comes from a real inner product space [Sch52]. Here we will prove the former result, that a real inner product space satisfies Ptolemy's inequality.

Proof. Note that to prove the inequality, one can take $V_0 = 0$ (by shifting) and assume the remaining vectors are non-zero (since otherwise the inequality holds trivially). Under these assumptions, we aim to prove for all V_1, V_2, V_3 non-zero in a real inner product space,

$$\|V_3\| \|V_1 - V_2\| \leq \|V_2\| \|V_1 - V_3\| + \|V_1\| \|V_2 - V_3\|. \quad (\text{eq:shifted-ptolemy}) \quad (201)$$

Let $\tilde{V}_1, \tilde{V}_2, \tilde{V}_3$ be rescaled versions of V_1, V_2, V_3 :

$$\tilde{V}_1 = \frac{V_1}{\|V_1\|^2}, \quad \tilde{V}_2 = \frac{V_2}{\|V_2\|^2}, \quad \tilde{V}_3 = \frac{V_3}{\|V_3\|^2}. \quad (202)$$

Under this rescaling, the pairwise distances are rescaled as

$$\|\tilde{V}_i - \tilde{V}_j\| = \sqrt{\frac{1}{\|V_i\|^2} + \frac{1}{\|V_j\|^2} - \frac{2|\langle V_i, V_j \rangle|^2}{\|V_i\|^2 \|V_j\|^2}} = \frac{\|V_i - V_j\|}{\|V_i\| \|V_j\|}. \quad (203)$$

Then, the triangle inequality for $\|\cdot\|$ applied to $\tilde{V}_1, \tilde{V}_2, \tilde{V}_3$,

$$\|\tilde{V}_1 - \tilde{V}_2\| \leq \|\tilde{V}_1 - \tilde{V}_3\| + \|\tilde{V}_2 - \tilde{V}_3\|, \quad (204)$$

is then equivalent to Eq. (201). $\square$

E.3.2 Cayley-Menger Determinant

First we note that the four Hermitian matrices $(V_0, V_1, V_2, V_3) = (0, A, C, E)$ necessarily lie inside a three-dimensional subspace of the set of $n \times n$ Hermitian matrices and thus there exists an orthonormal set of Hermitian matrices L_1, L_2, L_3 such that for each $i \in \{0, 1, 2, 3\}$

$$V_i = \sum_{j=1}^3 (v_i)_j L_j \quad (205)$$

In this basis, the vectors of coefficients $\{v_0, v_1, v_2, v_3\}$ constitute the vertices of a tetrahedra in $\mathbb{R}^3$ and moreover $\langle V_i, V_j \rangle = \text{Tr}(V_i^* V_j) = \langle v_i, v_j \rangle$. Following the wikipedia derivation, define a 4×4 matrix by

$$P = \begin{pmatrix} v_0 & v_1 & v_2 & v_3 \\ 1 & 1 & 1 & 1 \end{pmatrix} \quad (206)$$

such that $|\det(P)| = 3! \text{Vol}(v_0, v_1, v_2, v_3)$ where $\text{Vol}(v_0, v_1, v_2, v_3)$ is the volumes of the tetrahedron with vertices $v_0, v_1, v_2, v_3 \in \mathbb{R}^3$. Then one augments P to

$$\tilde{P} = \begin{pmatrix} v_0 & v_1 & v_2 & v_3 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ \|v_0\|^2 & \|v_1\|^2 & \|v_2\|^2 & \|v_3\|^2 & 1 \end{pmatrix} \quad (207)$$

such that $\det(P) = \det(\tilde{P})$ and also defines a 5×5 matrix Q by

$$Q = \begin{pmatrix} -2 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad (208)$$

Then $\tilde{P}^\dagger Q \tilde{P}$ is a 5×5 matrix

$$\tilde{\Sigma} = \tilde{P}^\dagger Q \tilde{P} = \begin{pmatrix} 0 & d_{01}^2 & d_{02}^2 & d_{03}^2 & 1 \\ d_{10}^2 & 0 & d_{12}^2 & d_{13}^2 & 1 \\ d_{20}^2 & d_{21}^2 & 0 & d_{23}^2 & 1 \\ d_{30}^2 & d_{31}^2 & d_{32}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix} \quad (209)$$

where

$$d_{ij}^2 := \|v_i - v_j\|^2 = \|v_i\|^2 + \|v_j\|^2 - 2 \langle v_i, v_j \rangle. \quad (210)$$

The determinant $\det(\tilde{\Sigma})$ is equal to

$$\det(\tilde{\Sigma}) = \det(\tilde{P}^\dagger Q \tilde{P}) = \det(Q) \det(P)^2 = 8 \cdot (3!)^2 \cdot \text{Vol}^2 \geq 0. \quad (211)$$

Using the isometric embedding of $(V_0, V_1, V_2, V_3) = (0, A, C, E)$ into $\mathbb{R}^3$, the Cayley-Menger determinant inequality takes the form

$$\det \begin{pmatrix} 0 & \|A\|^2 & \|C\|^2 & \|E\|^2 & 1 \\ \|A\|^2 & 0 & \|B\|^2 & \|F\|^2 & 1 \\ \|C\|^2 & \|B\|^2 & 0 & \|D\|^2 & 1 \\ \|E\|^2 & \|F\|^2 & \|D\|^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix} \geq 0 \quad (212)$$

E.4 Strongly subadditive/Submodular inequalities

DISCLAIMER FROM TC: The remainder of this section is written with respect to the A, B, C convention instead of the A, B, D convention. When discussing the tetrahedral Horn problem and its relation to the ordinary Horn problem or when matching the problem to the properties of the $6j$ symbols, the A, B, D convention is preferred. However when talking about submodular/strong subadditivity inequalities and comparing with the literature, the A, B, C convention is preferred. Before finalizing the paper, it will be desirable to establish a globally consistent convention. An equivalent formulation is to consider eigenvalues of four PSD matrices W, X, Y, Z with eigenvalues $(\omega, \chi, \nu, \zeta)$ such that $X + Y = W + Z$ and $W \geq X \geq Z$ and $W \geq Y \geq Z$.

Another strategy for understanding eigenvalues of positive semidefinite matrices and their sums is to consider the expansion of tensors powers of sums of matrices in relation to the tensor powers of their summands.¹¹ For instance, given a positive integer k and positive semidefinite A and B , using the non-commutative binomial expansion of $(A + B)^{\otimes k}$ and dropping all summands involving both A and B , one obtains [BS15, Proposition 1.1]:

$$(A + B)^{\otimes k} \geq A^{\otimes k} + B^{\otimes k}. \quad (213)$$

This technique, was generalized by [TCL11] [TCL11, Lemma 2.2] to obtain the following result.¹²

¹¹We will exploit a variation of this strategy repeatedly in Section 3.

¹²Further extensions of this technique include inequalities involving the sum $A + C$ which are outside the scope of the tetrahedral Horn problem (see [Lin14; BS15; NS23])

Proposition 54. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let k be a positive integer. Then*

$$(A + B + C)^{\otimes k} + B^{\otimes k} \geq (A + B)^{\otimes k} + (B + C)^{\otimes k}. \quad (214)$$

From Proposition 54 one can obtain an inequalities pertaining only to the eigenvalues of $B, A + B, B + C$, and $A + B + C$ by pairing the operator inequality with any $U(n)$ -equivariant positive semidefinite operator acting on $(\mathbb{C}^n)^{\otimes k}$.¹³

cor:schur_ineq

Corollary 55. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let s_λ be the Schur polynomial indexed by the Young diagram $\lambda' = (\lambda'_1 \geq \lambda'_2 \geq \dots \geq \lambda'_n)$ with size $|\lambda'| := \lambda'_1 + \dots + \lambda'_n = k$. Then*

$$s_\lambda(\text{eig}(A + B + C)) + s_\lambda(\text{eig}(B)) \geq s_\lambda(\text{eig}(A + B)) + s_\lambda(\text{eig}(B + C)). \quad (215)$$

{eq:schur_ineq}

Proof. Given a Young diagram λ' , let $P_{\lambda'}$ be the Hermitian projection operator onto the isotypic subspace $V_{\lambda'}^n \otimes W_{\lambda'} \subseteq (\mathbb{C}^d)^{\otimes k}$ (see Section 3 for details). Given positive semidefinite matrix $X \geq 0$, one can show that

$$\text{Tr}(P_{\lambda'} X^{\otimes k}) = \text{Tr}(P_{\lambda'} \text{diag}(\text{eig}(X))^{\otimes k}) = \dim(W_{\lambda'}) s_\lambda(\text{eig}(X)) \geq 0. \quad (216)$$

Now we apply the positivity preserving map $\text{Tr}(P_{\lambda'} \cdot)$ to Proposition 54 to obtain

$$\text{Tr}[P_{\lambda'}(A + B + C)^{\otimes k}] + \text{Tr}[P_{\lambda'} B^{\otimes k}] \geq \text{Tr}[P_{\lambda'}(A + B)^{\otimes k}] + \text{Tr}[P_{\lambda'}(B + C)^{\otimes k}] \quad (217)$$

and since λ' has at most n rows, we conclude $\dim(W_{\lambda'}) \neq 0$ and thus Eq. (215) holds. $\square$

In the special where where $\lambda' = (\underbrace{1, \dots, 1}_n)$, where $\text{Tr}(P_{(\underbrace{1, \dots, 1}_n)} X^{\otimes n}) = \det(X)$ one obtains the well-known determinant inequality [TCL11, Lemma 2.2]:

$$\det(A + B + C) + \det(B) \geq \det(A + B) + \det(B + C). \quad (218)$$

{eq:lin_det}

Another well-known determinant inequality is the multiplicative variant of Eq. (218) which states that for all $A, B, C \geq 0$ positive semidefinite,

$$\det(A + B + C) \det(B) \leq \det(A + B) \det(B + C). \quad (219)$$

{eq:mul_det}

This inequality follows as a special case of a general result due to Sagnol [Sag13, Proposition 2.3].

thm:sagnol_ineq

Theorem 56. *Let $A, B, C \geq 0$ be positive semidefinite matrices. Let f be a real valued and differentiable function over the non-negative real numbers $\mathbb{R}_{\geq 0}$ whose derivative f' is an operator monotone function over the positive real numbers $\mathbb{R}_{> 0}$. Then the following inequality holds*

$$\text{Tr } f(A + B + C) + \text{Tr } f(B) \geq \text{Tr } f(A + B) + \text{Tr } f(B + C). \quad (220)$$

{eq:sagnol_ineq}

Proof. Here we sketch the proof in [Sag13, Section 2.1]. Let $F = \text{Tr} \circ f$. The claim then amounts to proving for all $A, B, C \geq 0$,

$$\Delta_A \Delta_C F(B) = F(A + B + C) - F(A + B) - F(B + C) + F(B) \geq 0, \quad (221)$$

{eq:sagnol_diff}

where Δ_X is the difference operator $[\Delta_X F](Y) := F(X + Y) - F(Y)$. By standard continuity arguments (and the assumption that f is continuous), we may assume everything is positive definite in which case

$$\partial_A \Delta_C F(B) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \Delta_{\varepsilon A} \Delta_C F(B) \quad (222)$$

$$= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (\Delta_C F(B + \varepsilon A) - \Delta_C F(B)) \quad (223)$$

$$= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (F(\varepsilon A + B + C) - F(\varepsilon A + B) - F(B + C) + F(B)) \quad (224)$$

$$= \text{Tr} [(f'(B + C) - f'(B))A]. \quad (225)$$

{eq:sagnol_final_step}

¹³Paksoy, Turkmen, and Zhang has used a similar approach to prove analogous inequalities concerning so-called generalized matrix functions d_χ^H associated to irreducible characters χ of subgroups H of the symmetric group S_k [MM65; HM21]. Unfortunately, these generalized matrix functions are generally not invariant under unitary conjugation and thus depend on more than just eigenvalues; the only exception is the determinant which is already a special case of Corollary 55.

Using the assumption that f' is an operator monotone (and the assumption of positive definiteness) together with $B + C \geq B$ (because $C \geq 0$), we have $f'(B + C) \geq f'(B)$ and thus $f'(B + C) - f'(B) \geq 0$ is a positive semidefinite operator. Therefore, the function $\Delta_C F(B)$ is non-decreasing under $B \mapsto A + B$ and thus Eq. (221) holds. $\square$

As immediate corollaries to Theorem 56, letting $f(x) = \log x$ and noting $\text{Tr}(\log X) = \log \det(X)$, one recovers Eq. (219). Moreover, taking $f(x) = -x^p$ for $p \in [0, 1]$, one obtains [BGS08, Lemma 5.3] which states for all $A, B, C \geq 0$,

$$\forall p \in [0, 1] : \sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \geq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (\text{eq:bouhtou_ineq}) \quad (226)$$

In the limit where $p \rightarrow 0^+$, Sagnol also obtains at first order the well-known rank inequality [Sag13, Proposition A.1]

$$\text{rank}(A + B) + \text{rank}(B + C) \geq \text{rank}(B) + \text{rank}(A + B + C). \quad (227)$$

On the other hand, letting $f(x) = x^p$ for $p \in (1, 2]$ yields for all $A, B, C \geq 0$

$$\forall p \in (1, 2] : \sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \leq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (\text{eq:bouhtou_ineq_reverse}) \quad (228)$$

Of particular interest to us in Section 4 is the function $f(x) = x \log x$ which implies for all $A, B, C \geq 0$ the inequality

$$S(A + B) + S(B + C) \geq S(B) + S(A + B + C) \quad (\text{eq:ssa_like_ineq}) \quad (229)$$

where $S(X) = s(\text{eig}(X)) := -\sum_{i=1}^n \lambda_i(X) \log \lambda_i(X)$ is the *unnormalized* von Neumann entropy of X . Note that a similar inequality involving entropies of sums of positive semidefinite matrices (subject to additional eigenvalue constraints) has appeared in [CŠW18]. Moreover, while it can be shown that Eq. (229) is strictly tighter than the inequality in [CŠW18] for fixed matrices, it is also possible to obtain Eq. (229) from the result in [CŠW18] by downscaling the involved matrices in the sense of Proposition 53.

Remark 57. One consequence of Theorem 52, which relates the eigenvalues of sums of positive semidefinite matrices to the eigenvalues of diagonal blocks of larger positive semidefinite matrices with bounded rank, is that many of the aforementioned inequalities can now be seen as consequences of more general inequalities which constrain the eigenvalues of blocks.

For instance, the Kotljanskii determinant inequality [JB93] (in the notation of Theorem 52) implies for any 3×3 block matrix, that

$$\det(C') \det(F') \geq \det(E') \det(B'). \quad (230)$$

Using Theorem 52, the Kotljanskii determinant inequality becomes Eq. (219). More generally, Audenaert, Hiai, and Petz were interested in functions f which are strongly superadditive in the sense that for any 3×3 block matrices E' (again in the notation of Theorem 52) the following inequality holds [AHP10]:

$$\text{Tr } f(E') + \text{Tr } f(B') \geq \text{Tr } f(C') + \text{Tr } f(F'). \quad (\text{eq:audenaert}) \quad (231)$$

Their conclusion, also discovered by Friedland and Gaubert [FG13] was that it is *sufficient* for f' to be an operator monotone to conclude Eq. (231) holds. In light of Theorem 52, the results of Audenaert, Hiai, and Petz and [FG13] give an alternative proof of Theorem 56. In [LS14], Lewin and Sabin further proved that f' being an operator monotone is also *necessary* for Eq. (231) to hold for all 3×3 block matrices E' . Note that from the proof of Theorem 56 above, it is not too difficult to see that f' being an operator monotone is also necessary for Eq. (220) to hold (otherwise it becomes possible for Eq. (225) to become negative).

Remark 58. Much of the interest in characterizing the eigenvalues of principal submatrices of positive semidefinite matrices is closely related to the correspondence between positive semidefinite matrices and covariance matrices of Gaussian multivariate random variables [Lam+17]. From this perspective, principal submatrices become covariance matrices of Gaussian marginal distributions, and thus one may view the principal submatrix eigenvalue problem as a kind of “spectral” classical Gaussian marginal problem. Note that from this perspective, the differential entropies of the underlying Gaussian distributions are equal to logarithms of determinants up to an additive constants [Lam+17]. The Gaussian entropy region has been partially characterized by Shadbakht and Hassibi [SH11; HS08].

E.5 The Courtade-Sra determinant inequality

Many of the previously discussed inequalities constrain the eigenvalues of positive semidefinite matrices $B, A+B, B+C$ and $A+B+C$. Here we show how to improve the determinant inequality $\det(A+B)\det(B+C) \geq \det(A+B+C)\det(B)$ in a manner which also accounts for the determinants $\det(A)$ and $\det(C)$. This inequality exploits Minkowski's determinant inequality for positive semidefinite $n \times n$ matrices A and B which reads (see [HJ85, Theorem 7.8.21] or [Bha97, Equation II.44]):

$$\det(A+B)^{\frac{1}{n}} \geq \det(A)^{\frac{1}{n}} + \det(B)^{\frac{1}{n}}. \quad (232)$$

The following inequality can be attributed to an information-theoretic result obtained by Courtade [Cou16]¹⁴, but the elementary proof we present here is do actually due to Sra [Sra]. This inequality also appears in the recent review paper by Niculescu and Sra [NS23, Section 6.2].

prop:cs_ineq

Proposition 59 (Courtade-Sra determinant inequality). *Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices. Then*

$$\det(A+B)^{\frac{1}{n}} \det(B+C)^{\frac{1}{n}} \geq \det(A+B+C)^{\frac{1}{n}} \det(B)^{\frac{1}{n}} + \det(A)^{\frac{1}{n}} \det(C)^{\frac{1}{n}}. \quad (233)$$

{eq:cs_ineq}

Proof. By standard continuity arguments, it suffices to prove the inequality for $B > 0$ positive definite (if it holds for positive definite $B + \varepsilon I$, then continuity permits taking the limit $\varepsilon \rightarrow 0$). As B is positive definite, it is also invertible and we may define

$$X = B^{-\frac{1}{2}}AB^{-\frac{1}{2}}, \quad Y = B^{-\frac{1}{2}}CB^{-\frac{1}{2}}. \quad (234)$$

By dividing Eq. (233) by $\det(B)^{\frac{2}{n}} \neq 0$, it then suffices to prove

$$\det(I+X)^{\frac{1}{n}} \det(I+Y)^{\frac{1}{n}} \geq \det(I+X+Y)^{\frac{1}{n}} + \det(X)^{\frac{1}{n}} \det(Y)^{\frac{1}{n}}. \quad (235)$$

{eq:cs_ineq_xy}

Let $\lambda(X) = \text{eig}(X)$ and $\lambda(Y) = \text{eig}(Y)$. Then for any permutation $\pi \in S_n$ let

$$L_\pi = \text{diag}\{1 + \lambda_i(X) + \lambda_{\pi(i)}(Y)\}_{i=1}^n, \quad R_\pi = \text{diag}\{\lambda_i(X)\lambda_{\pi(i)}(Y)\}_{i=1}^n. \quad (236)$$

Then we have the identity

$$\det(I+X)\det(I+Y) = \prod_{i=1}^n (1 + \lambda_i(X)) \prod_{i=1}^n (1 + \lambda_i(Y)), \quad (237)$$

$$= \prod_{i=1}^n (1 + \lambda_i(X) + \lambda_{\pi(i)}(Y) + \lambda_i(X)\lambda_{\pi(i)}(Y)), \quad (238)$$

$$= \det(L_\pi + R_\pi) \quad (239)$$

Here we can apply Minkowski's theorem to obtain

$$\det(L_\pi + R_\pi)^{\frac{1}{n}} \geq \det(L_\pi)^{\frac{1}{n}} + \det(R_\pi)^{\frac{1}{n}}. \quad (240)$$

By definition, we have $\det(R_\pi) = \det(X)\det(Y)$ for any permutation π , while a theorem do to Fiedler [Fie71] states (see also [Bha97, Exercise VI.7.2, Theorem VI.7.1])

$$\det(I+X+Y) \leq \max_{\pi} \prod_{i=1}^n (\lambda_i(I+X) + \lambda_{\pi(i)}(Y)) = \max_{\pi} \det(L_\pi). \quad (241)$$

Combining these bounds together yields Eq. (235) and thus the claim holds. $\square$

¹⁴Note in [Cou17], Courtade conjectures a generalization of the Brunn-Minkowski inequality for bounded convex sets A, B, C in $\mathbb{R}^n$ of the form $V(A+B)^{\frac{1}{n}}V(B+C)^{\frac{1}{n}} \geq V(A+B+C)^{\frac{1}{n}}V(B)^{\frac{1}{n}} + V(A)^{\frac{1}{n}}V(C)^{\frac{1}{n}}$. See also [CC84].

TC: On mathoverflow <https://mathoverflow.net/questions/257562/a-determinant-inequality>
Sra also proves

$$E_k(B)E_k(A+B+C) \leq E_k(A+B)E_k(B+C) \quad (242)$$

where $E_k(X) = e_k(\text{eig}(X))$ where e_k is the k th elementary symmetric polynomial. TC: I conjecture that Eq. (233) also holds for more general hyperbolic polynomials, although I do not have a proof yet:

Conjecture 60. *Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices. Then*

$$E_k^{\frac{1}{k}}(A+B)E_k^{\frac{1}{k}}(B+C) \geq E_k^{\frac{1}{k}}(A+B+C)E_k^{\frac{1}{k}}(B) + E_k^{\frac{1}{k}}(A)E_k^{\frac{1}{k}}(C), \quad \{\text{eq:cs_ineq_conjecture}\} \quad (243)$$

where $E_k(X) = e_k(\text{eig}(X))$ and where e_k is the k th elementary symmetric polynomial.

Proving this conjecture has proven quite challenging.

What I believe I'm managed to prove is the following.

Let $\phi : \text{PSD}_n \rightarrow \mathbb{R}$ be a function on the cone of positive semidefinite matrices. If ϕ is such that $\log \phi$ is 2-monotone, i.e.,

$$-(\partial_A \log \phi)(B) \geq 0, \quad +(\partial_C \partial_A \log \phi)(B) \geq 0, \quad (244)$$

then it is fairly clear that

$$\log \phi(A+B) + \log \phi(B+C) \geq \log \phi(B) + \log \phi(A+B+C), \quad (245)$$

or equivalently,

$$\phi(A+B)\phi(B+C) \geq \phi(B)\phi(A+B+C). \quad (246)$$

If in addition to $\log \phi$ being 2-monotone, ϕ satisfies a kind of 3-montonicity, namely

$$-\partial_W(\phi^2 \partial_C \partial_A \log \phi)(B) \geq 0, \quad (247)$$

then,

$$[\phi(A+B+W)\phi(B+C+W) - \phi(A+B+C+W)\phi(B+W)] - [\phi(A+B)\phi(B+C) - \phi(A+B+C)\phi(B)] \geq 0 \quad (248)$$

which means, assuming $\phi(0) = 0$ and for $B \mapsto 0$ and $W \mapsto B$, we have desired inequality:

$$\phi(A+B)\phi(B+C) - \phi(A+B+C)\phi(B) - \phi(A)\phi(C) \geq 0. \quad (249)$$

Then what remains is to prove that $\phi = E_k^{\frac{1}{k}}$ has this property.

TC: References that might be useful: The strongly subadditive inequality relies on the log concavity result for hyperbolic polynomials due to Bauschke et al. [Bau+01]. An excellent crash course into Gårding's theory of hyperbolic polynomials (as well as some important results) is also [Bau+01]. For the particular case of the real vector space of $n \times n$ Hermitian matrices of dimension $n(n+1)/2$, the polynomial $X \mapsto \det(X)$ is hyperbolic in the direction of the identity matrix id_n (i.e., Hermitian matrices only have real eigenvalues). Moreover, the homogeneous polynomial $p_m(X) = \text{Tr}(P_{(1,m,1)} X^{\otimes m})$ is the sum of the principal minors of X of size m and also equals the elementary symmetric polynomial of the eigenvalues of X , i.e.,

$$p_m(X) = E_m(\text{eig}(X)). \quad (250)$$

The homogeneous polynomial $p_m(X)$, like the determinant $\det(X) = p_n(X)$ itself, is also hyperbolic in the direction of the identity matrix id_n . In this context, together with the positivity $p_m(\text{id}_n) > 0$, Gårding's inequality generalizes the Minkowski determinant inequality and yields superlinearity of $p_m(X)^{\frac{1}{m}}$:

$$p_m(X+Y)^{\frac{1}{m}} \geq p_m(X)^{\frac{1}{m}} + p_m(Y)^{\frac{1}{m}} \quad (251)$$

for all X, Y positive semidefinite (the hyperbolicity cone for p_m). This inequality is due to Marcus and Lopes [ML57, Theorem 2], but it can also be proven using joint concavity of $(A_1, \dots, A_m) \mapsto A_1^{\frac{1}{m}} \otimes \dots \otimes A_m^{\frac{1}{m}}$ and thus concavity of $(A^{\frac{1}{m}})^{\otimes m}$ and thus by homogeneity the operator inequality $((A+B)^{\frac{1}{m}})^{\otimes m} \geq (A^{\frac{1}{m}})^{\otimes m} + (B^{\frac{1}{m}})^{\otimes m}$ ([Bha09, Exercise 4.2.9]).

See [ML57; McL59; AMT84] for using inequalities and proof techniques which might be relevant. See also [AMT84] or [Bha09, Chapter 4] for many properties of matrix means and their ability to prove minkowski-like inequalities. See also Khovanskii [Kho84] for inequalities mirroring the Aleksandrov-Fenchel type inequalities for hyperbolic polynomials (i.e., the mixed discriminant inequalities of Aleksandrov... see Shenfeld and Van Handel for proofs [SV19].) See [KMS19] for representation theorems and other results concerning complete monotonicity.

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